Linnik's constant below the zero-density exponent: 11/5 from a zero-free half-plane and the Chen–Gupta–Li density estimate

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Draft, 30 October 2026

Abstract

We prove that there is a constant K such that every reduced residue class modulo q contains a prime $p \le Kq^{11/5}$. The proof uses two recent results that have not been refereed: OpenAI's zero-free half-plane $\operatorname{Re} s > 11/12$ for Dirichlet L-functions, and Chen, Gupta and Li's zero-density estimate, which lowers the density exponent to $7/3$. From these the classical argument gives every exponent above $7/3$. We go below the density exponent with Harman's sieve, fed by a Type II estimate that sorts the characters by the real parts of the zeros of their L-functions: Cauchy–Schwarz over each class of characters halves the density exponent A, so that, for the primes up to $x = q^L$, the estimate holds for Dirichlet polynomials of length about q once L exceeds $1 + A/2$, which is $13/6$ for $A = 7/3$. A bilinear Type I estimate, proved by Parseval's identity, lets the sieve also decompose terms whose moduli exceed x/q, and the sieve's loss is bounded by a certificate computed in exact integer arithmetic. The whole proof is formalised in Lean 4 with Mathlib, including the half-plane and, from Chen, Gupta and Li's Proposition 3.1, the density estimates it uses, and depends only on Lean's standard axioms. The exponent is to be compared with $7/3$, not with Xylouris's 5, whose proof uses refereed results only.

1 Introduction

Theorem 1.1. *There is K such that for every $q \ge 1$ and every a prime to q there is a prime $p \equiv a \pmod q$ with $p \le Kq^{11/5}$.*

Linnik [1, 2] proved that there are absolute constants K and L such that every reduced residue class mod q contains a prime $p \le Kq^L$ (see [21,

Chapter 18] for an account). The admissible exponent L, Linnik's constant, has been lowered many times: Heath-Brown [3] reached $L = 5.5$, and Xylouris $L = 5.18$ [4] and then $L = 5$ [5], and again $L = 5$ in a short version, which notes that a slightly smaller value follows [6]. Under the generalised Riemann hypothesis every $L > 2$ is admissible [7, 8]. Theorem 1.1 has $L = 11/5 = 2.2$, but it is not on the same footing as these exponents, whose proofs use refereed results only: its proof uses two recent results that have not been refereed (Section 2), and with them the classical argument already gives every $L > 7/3$. What this paper adds is measured from $7/3$.

The zeros near $\operatorname{Re} s = 1$

The difficulty in Linnik's theorem is the zeros of the L-functions mod q near the line $\operatorname{Re} s = 1$. By the explicit formula, a zero $\rho = \beta + i\gamma$ of $L(s, \chi)$ contributes about $x^{\beta-1}$, relative to the main term, to the count of primes $p \le x$ in a class mod q. A zero-density estimate with exponent A, a bound of about $(qT)^{A(1-\sigma)}$ for the number of zeros with real part at least σ, makes the zeros of real part about σ contribute about $(q^A/x)^{1-\sigma}$ together. Classically the zero-free region allows $1-\beta$ as small as $c/\log q$, and one exceptional real zero may be closer still; at $x = q^L$ such a zero contributes a constant, not a small quantity. The record proofs control these zeros with zero-free regions, density estimates without logarithmic factors and the Deuring–Heilbronn phenomenon, and the constants of all three enter the exponent.

The zero-free half-plane of OpenAI's release [9, 10], Input 1, removes the zeros with $\beta > 11/12$. Every remaining zero then contributes at most a power $x^{-1/12}$, and the problem becomes one of density: a density exponent $A \ge 2$ for $1/2 \le \sigma \le 11/12$ gives Linnik's theorem with every exponent $L > A$. Huxley's density estimate [25] gives $A = 12/5$. Chen, Gupta and Li [12], whose method is Guth and Maynard's [13], prove $A = 7/3$, uniformly in q and T: their Theorem 1.2 bounds the count, for a divisor q_1 of q, by four terms, and at $q_1 = q$ the largest of them on $[7/10, 4/5]$ has the exponent $4/(1+\sigma)$, which is $7/3$ at $\sigma = 5/7$ and smaller beyond; below $5/7$ Ingham's exponent $3/(2-\sigma)$ is at most $7/3$. With Input 1 the classical deduction turns the density exponent $7/3$ into every Linnik exponent above $7/3$ (Section 8). Neither that deduction nor the exponent of this paper needs the value $11/12$. The deduction works with any zero-free half-plane $\operatorname{Re} s > 1 - \eta_0$, for a fixed $\eta_0 > 0$, and so does the sieve: with the paper's margins its window stays admissible for every $\eta_0 \ge 2.85 \cdot 10^{-4}$, as checked at the corners of the window, and with smaller margins for any fixed $\eta_0 > 0$; neither remark is formalised (Section 8). What both use is that there are no zeros near $\operatorname{Re} s = 1$, and in

particular no exceptional zero.

Why a sieve can go below the density exponent

For primes in short intervals a density exponent A gives intervals of length $x^{1-1/A}$, and Harman's sieve goes further, to $x^{0.525}$ [41]. The same happens here. The sieve needs a *Type II estimate*: a bound for bilinear sums $\sum_\chi |W(\chi)||S(\chi)|$ over the non-trivial characters mod q, with W a Dirichlet polynomial of length $P = q^\mu$ with bounded coefficients and S a product of prime sums of total length x/P. Sort the characters by the largest real part β of the zeros of their L-functions. Each prime sum of length Q is then at most about Q^β, so S is at most about $(x/P)^\beta$, and a density estimate with exponent A bounds the number of characters by $q^{A(1-\beta)}$. Cauchy–Schwarz over these characters and orthogonality bound $\sum |W(\chi)|$ by the square root of that number times $((q+P)P)^{1/2}$, and the square root is the gain: it halves the density exponent. Neglecting logarithms and margins, this bound is smaller than the main term for the lengths P around q exactly when L exceeds $1+A/2$ for the largest density exponent A (Section 3.5). With $A = 7/3$ this threshold is $13/6 \approx 2.167$, against $7/3$ for the classical argument.

The sieve needs a window of positive width, and it pays for the sums it cannot compare. At $L = 11/5$ the window is $[u_1, u_2] = [0.43457, 0.46959]$ in exponents of $x = q^L$ (Section 3), and the primes that the sieve loses, bounded by a finite computation, are at most 0.9331 of them (Section 5). The method has a little room below $11/5$: with the window chosen afresh it reaches about 2.198, by a computation that is not formalised (§S6), and $11/5$ is the round exponent that the paper proves. The table compares what the inputs give with and without the sieve (Section 8).

Inputs	Method	Exponent
Input 1	density deduction	$12 + \varepsilon$
Input 1, Ingham	density deduction	$36/13 + \varepsilon \approx 2.769$
Input 1, Huxley	density deduction	$12/5 + \varepsilon = 2.4 + \varepsilon$
Inputs 1 and 2, the curve g^*	density deduction	$2.3334 + \varepsilon$
Input 1, Chen–Gupta–Li's estimate [12]	density deduction (not formalised)	$7/3 + \varepsilon \approx 2.333$
Input 1, Ingham, Huxley, in steps	sieve, computed (not formalised)	$2239/1000 = 2.239$
Inputs 1 and 2, the curve g^*	sieve, this paper	$11/5 = 2.2$

The gain does not depend on Chen, Gupta and Li's estimate: with the half-plane and the classical estimates of Ingham and Huxley alone, the same sieve gives about 2.24 in place of $12/5$, by a computation that is not formalised.

Harman's sieve, and what is new

Harman's sieve [38], [39, Chapter 3] bounds from below the number of primes in the set $\mathcal{A}$ of integers $n \in (x, 2x]$ with $n \equiv a \pmod q$ by comparing it with the set $\mathcal{B}$ of the integers $n \in (x, 2x]$ prime to q, weighted by $1/\varphi(q)$. It needs two kinds of information. Type I information is the comparison of $\mathcal{A}$ with $\mathcal{B}$ on the multiples of each integer prime to q, which counting in residue classes gives. Type II information is the comparison on bilinear sums $\sum_{mn} a_m b_n$, with m in the window's range $[x^{u_1}, x^{u_2}]$, bounded coefficients a_m, and b_n supported on products of primes, which the Type II estimate gives. The sieve recovers the primes from the sums it can compare, and a finite computation bounds the fraction it loses.

The paper's contributions are three.

1. *A Type II estimate over the zeros' real parts* (Section 3). The characters mod q are sorted into bins by the largest real part of the zeros of their L-functions up to a height. In each bin the bilinear sum is bounded by the zeros, which make the Dirichlet polynomial of the primes small, and by the number of characters in the bin, which a density estimate bounds; the better of orthogonality and the Halász–Montgomery inequality is taken in each bin. Inputs 1 and 2, with the classical density estimates of Ingham and Huxley, give the window $[u_1, u_2] = [43457/100000,\ 46959/100000]$ at $L = 11/5$.

2. *A decomposition by Buchstab's identity to depth eight, with Type I information beyond x/q* (Sections 4 and 5). The identity is applied to depth eight, the primes of the four deepest levels on a coarser grid. The terms in which the product of some of the prime variables has its exponent in the window, or in its reflection $[1 - u_2, 1 - u_1]$, are Type II sums and are evaluated; of the others, a term with sign $-$ is decomposed further, and a term with sign $+$ is dropped or decomposed further. Classically a term can be decomposed only while the product of its primes stays below x^{1-u_1}, its part above $X' = x^{1-u_2}$ being a Type II sum; beyond, the cofactor falls below the window, and the term could only be dropped, at a cost which at $L = 11/5$ would exceed all the primes. Such a term is now decomposed whole when its primes, with the smaller primes still to come, split into two groups $d_1 < x/q$ and d_2 with $d_1 d_2^2 < x$, which reaches moduli up to about $x/q^{1/2}$: the leaves of its tree are then evaluated by a bilinear Type I estimate, proved by Parseval's identity, with no bound for Kloosterman sums. With the split alone the certificate's total would be 2.43, against 0.932

with the bilinear estimate (Section 5).

3. *A certificate* (Section 5). The fraction of the primes the decomposition loses is bounded above by a sum over the cells of a grid of the exponents, computed exactly in integers, with Buchstab's function itself where its argument lies in $[2, 3]$ and the trapezoid rule for the cells' prime sums. On a grid of 272 cells a side (and 68 cells of four for the deepest primes) the sum is 0.93214, and Lean's kernel checks that it is below 0.93308, so below 1: a positive proportion of the primes remains.

The rest is not new. Inputs 1 and 2 are OpenAI's and Chen, Gupta and Li's, and the density estimates of Ingham [17] and Huxley [24, 25], the Halász–Montgomery inequality [27, 28], Buchstab's identity [45], Brun's sieve [44] and Harman's sieve are classical.

Status and formalisation

The Lean statement of Theorem 1.1 has no hypotheses, and its axioms are Lean's standard three, so the theorem is unconditional. Its proof uses two recent results that have not been refereed, OpenAI's zero-free half-plane and Chen, Gupta and Li's zero-density estimate, which Section 2 states as Inputs 1 and 2. The library proves both, the half-plane along OpenAI's companion preprint and the density estimate from Chen, Gupta and Li's Proposition 3.1, so the truth of the theorem rests on Lean's kernel, Mathlib and the pinned toolchain (with the vendored upstream files and the provenance of Mathlib's prebuilt files, §S5), not on the preprints. Whether the Lean statements of the inputs faithfully capture the preprints' theorems matters for attributing and interpreting them, not for the truth of the theorem, and the kernel does not check it. Every step of the proof, and every result from the literature that it uses, is proved in the Lean library [49], written with Lean 4 [47] and Mathlib [48], or in Mathlib itself, which supplies among others Dirichlet's theorem and the functional equation of Dirichlet L-functions. Appendix A names the theorems, Section 7 says what is formalised and what is known of independent checking of the inputs, and the directory `formalisations/` of the repository documents each result from the literature: its source, its Lean statement and the differences between them. The proofs of the paper are those of the library, written at the level of detail of a paper; where the Lean proof fixes a constant whose value the argument does not need, the paper usually says only that the constant exists. The argument, the formalisation and this text were developed with the help of

AI systems (see the acknowledgements); the author is responsible for the paper.

Related work

Heath-Brown and Iwaniec [40] combined a sieve with zero-density estimates, which bound Dirichlet polynomials over primes in the t-aspect; the bins of Section 3 are the analogue in the q-aspect, over the characters mod q. Harman's sieve with a numerical bound for its loss is the method of Baker, Harman and Pintz [41] and of R. Li [42]. R. Li also adapts his calculation to moduli all of whose prime factors are below q^C, under a zero-free region of width $(\log q)^{-3/4}$, and obtains $1/0.4752 < 2.1044$ for the powers of a fixed prime; Input 1 implies his hypothesis, so for such moduli his theorem, if it holds as stated, gives a smaller exponent (Section 8). Theorem 1.1 holds for every modulus.

Outline

Section 2 states the two inputs and the density curve. Section 3 proves the Type II estimate and explains where the window comes from, and Section 4 gives the Type I estimates, the classical one and a bilinear one, and the tools of Harman's sieve. Section 5 describes the decomposition, the cost of the terms it drops and its certificate, and Section 6 assembles the proof. Section 7 describes the formalisation, and Section 8 compares Theorem 1.1 with what the inputs give without the sieve, and discusses the method's limits. Appendix A names the Lean theorem behind each claim. A supplement holds the proofs of four standard results, of the kernel bound, and of Lemma 4.3 and Proposition 4.8 (§S1 to §S4), the Lean details of the paper (§S5) and the computations around the exponent (§S6).

2 Notation and the two inputs

Throughout, $q \geq 1$ is the modulus and a is prime to q. The Dirichlet characters mod q are χ, the trivial one is χ_0, and $L(s,\chi)$ is the L-function of χ, continued to the whole plane. For $1/2 \leq \sigma \leq 1$ and $T \geq 2$ let

$$N^*(\sigma, T; q) = \#\{\chi \bmod q : L(s,\chi) = 0 \text{ for some } s \in \mathcal{R}(\sigma, T)\},$$

where $\mathcal{R}(\sigma, T) = \{s : \operatorname{Re} s \geq \sigma,\ |\operatorname{Im} s| \leq T\}$: the number of characters mod q whose L-function has a zero in the rectangle. It counts characters, not zeros, so a bound for the number of zeros bounds it. The proof is a sieve

on the integers of $(x, 2x]$, $x = q^L$, with $L = 11/5$ in the end, and q is taken large.

The paper states two inputs, written *Input* and not *hypothesis* because the library proves them (Section 7). A statement below that holds 'under' an input is conditional in the text of the paper.

Input 1 (the zero-free half-plane)**.** For every $q \geq 1$, every Dirichlet character χ mod q and every s with $\operatorname{Re} s > 11/12$, $L(s, \chi) \neq 0$, except at $s = 1$ for the trivial character.

The exception is the pole of $L(s, \chi_0)$ at $s = 1$. This is OpenAI's result: their companion preprint [10] proves it, and their main preprint [9] proves the stronger $\operatorname{Re} s > 7/8$. The library proves it along the companion preprint's route (§S5).

Input 2 (zero-density estimates from 5/7 to 131/168)**.** Let $\sigma_k = 5/7 + k/1680$ for $0 \leq k \leq 14$ and $\sigma_k = \sigma_{14} + (k - 14)/280$ for $14 \leq k \leq 30$, so that $\sigma_{30} = 131/168$, and let $A_0, \ldots, A_{29}$ be the numbers of the table below. For each $k < 30$ there is C with the following property: for every $q \geq 1$, every $\sigma_k \leq \sigma \leq \sigma_{k+1}$ and every $T \geq 2$, at most $C(qT)^{A_k(1-\sigma)}(\log qT)^{10}$ characters mod q have an L-function with a zero in $\operatorname{Re} s \geq \sigma$, $|\operatorname{Im} s| \leq T$.

The A_k are multiples of 10^{-5} between $4/(1 + \sigma_k) + 3.5 \cdot 10^{-5}$ and $4/(1 + \sigma_k) + 8 \cdot 10^{-5}$:

$k = 0$–5 :	2.33340	2.33258	2.33178	2.33096	2.33016	2.32934
$k = 6$–11 :	2.32854	2.32772	2.32692	2.32610	2.32532	2.32452
$k = 12$–17 :	2.32370	2.32290	2.32212	2.31728	2.31252	2.30774
$k = 18$–23 :	2.30300	2.29826	2.29358	2.28889	2.28423	2.27959
$k = 24$–29 :	2.27493	2.27035	2.26575	2.26117	2.25661	2.25207

The exponent $4/(1 + \sigma)$ is that of the first term of Chen, Gupta and Li's Theorem 1.2 [12] at $q_1 = q$, the largest of its four terms on $[7/10, 4/5]$. It decreases in σ, so their theorem gives the bound of Input 2 on each step for every exponent above $4/(1+\sigma_k)$, the A_k among them; it counts zeros, which gives the count of characters stated here. Proposition 3.6 holds also with every A_k raised to $4/(1 + \sigma_k) + 10^{-4}$. The library derives each step from their Proposition 3.1, the large-values estimate, in the form for one point per character that their proof uses (distinct primitive characters, each with one height, and sums over $[6N/5, 9N/5]$ with $N \geq (qT)^{2/3}/2$ and $7/10 \leq \sigma \leq 4/5$); it does not prove their Theorem 1.2 as they state it (§S5).

The density curve

The Type II estimate of Section 3 needs a bound for $N^*(\sigma, T; q)$ at every σ in $[1/2, 1]$. It takes it in the form of a curve.

Definition 2.1. A function $g : \mathbb{R} \to \mathbb{R}$ is a *density curve* if there is $C \geq 0$ such that for every $q \geq 1$, every $1/2 \leq \sigma \leq 1$ and every $T \geq 2$,

$$N^*(\sigma, T; q) \leq C\,(qT)^{g(\sigma)}\big(1 + \log qT\big)^{10}.$$

The curve of this paper takes the trivial count below $4/7$, Ingham's $3/(2-\sigma)$ up to $5/7$, Chen, Gupta and Li's $4/(1+\sigma)$ up to $131/168$, Huxley's $3/(3\sigma-1)$ up to $5/6$, and a constant exponent above, where Input 1 removes the zeros beyond $11/12$. The window condition of Section 3 is checked piece by piece, on pieces of the form $A(1-\sigma)$, so the exponents are frozen on short steps and rounded up. The curve g^* is the following.

- $g^*(\sigma) = 1$ for $\sigma < 4/7$.
- On $[4/7, 5/7)$, divided into twenty steps $[\sigma_-, \sigma_+)$ of length $1/140$, $g^*(\sigma) = \big(3/(2-\sigma_+) + 1/2000\big)(1-\sigma)$: Ingham's exponent at the step's right end plus an excess of $1/2000$, except on the last step $[99/140, 5/7)$, where the excess is $1/100000$ and the exponent is $7/3 + 1/100000$.
- On $[5/7, 131/168)$, divided into the 30 steps $[\sigma_k, \sigma_{k+1})$ of Input 2, $g^*(\sigma) = A_k(1-\sigma)$.
- On $[131/168, 5/6)$, divided into fifteen steps $[\sigma_-, \sigma_+)$ of length $1/280$, $g^*(\sigma) = \big(3/(3\sigma_- - 1) + 1/2000\big)(1-\sigma)$: Huxley's exponent at the step's left end plus $1/2000$.
- $g^*(\sigma) = (101/50)(1-\sigma)$ for $\sigma \geq 5/6$.

The exponent rises from about 2.111 to $7/3+1/100000$ on $[4/7, 5/7]$, is 2.3334 on the first step after $5/7$, and then falls, to about 2.015 on the last step before $5/6$; from $5/6$ on it is $101/50$. On each step $A(1-\sigma) \leq A(1-\sigma_-) \leq 1$, so $g^* \leq 1$ everywhere, and $g^*(1/2) = 1$.

Proposition 2.2. *Under Inputs 1 and 2, g^* is a density curve.*

Proof. A finite union of ranges on each of which the bound holds, with its own constant, is a range on which it holds with the sum of the constants, so it suffices to prove the bound on each piece.

- For $\sigma < 4/7$: $N^*(\sigma, T; q) \leq \varphi(q) \leq qT$.

- On each step of $[4/7, 5/7]$: Ingham's estimate [17] in the form over the characters mod q, which bounds $N^*(\sigma, T; q)$ by $C(qT)^{A(1-\sigma)}(\log qT)^{10}$ on $[\sigma_-, \sigma_+]$ for every A above $3/(2-\sigma_+)$. Ingham proved his estimate for the zeros of ζ; the form over the characters is Theorem 12.1 of Montgomery's book [18] (as cited by Huxley [25]).

- On each step of $[5/7, 131/168]$: Input 2.

- On each step of $[131/168, 5/6]$: Huxley's estimate [25] over the characters mod q, with the exponent $3/(3\sigma_- - 1) + 1/2000$ on $[\sigma_-, \sigma_+]$. His (1.1) is a bound $(Q^2T/q_0)^{12(1-\sigma)/5+\varepsilon}$ for the zeros of the L-functions of the primitive characters to the moduli up to Q divisible by q_0, and his (2.23) has the exponent $3/(3\sigma - 1)$ there; with $Q = q_0 = d$ for each $d \mid q$, the case of one modulus, it bounds the characters mod q, each of which is induced by a primitive character mod some $d \mid q$ with the same zeros in $\operatorname{Re} s > 0$, at a cost $\tau(q) \ll q^{\varepsilon}$. The excess $1/2000$ absorbs this q^{ε} and Huxley's own ε, since $(1-\sigma)/2000 \geq 1/12000$ on these steps.

- For $\sigma \geq 5/6$: the exponent $101/50$. On $[5/6, 11/12]$ Huxley's exponent $3/(3\sigma - 1)$ is at most 2, so his estimate gives $2 + 1/50$, the excess $1/50$ absorbing the same two factors q^{ε}, since $(1-\sigma)/50 \geq 1/600$ there; above $11/12$ no character is counted, by Input 1.

Each of these bounds has a factor $(\log qT)^{10}$, which is at most $(1+\log qT)^{10}$. □

Two pieces of g^* are not classical. One is the steps of Input 2 on $[5/7, 131/168]$: their exponents start just above $7/3$ (2.3334) at $5/7$ and decrease like $4/(1+\sigma)$, and they replace the classical exponent $\min(3/(2-\sigma), 3/(3\sigma-1))$, which is Ingham's up to $\sigma = 3/4$, where both equal $12/5$, and Huxley's after. The other is the part above $11/12$, which Input 1 gives. Input 1 also enters the Type II estimate (Section 3) and, in the formalisation, the prime number theorem (Section 6).

3 The Type II estimate

Fix a smooth real function f_0 supported in $[1/2, 2]$. For a character χ, a scale $Q \geq 4$ and a height $h \in \mathbb{R}$, the smoothed prime sum is

$$S_Q(\chi, h) = \sum_{p \leq 2Q} \chi(p) \Big(\frac{p}{Q}\Big)^{-ih} f_0\Big(\frac{p}{Q}\Big),$$

and for a finite set $S \subseteq (P, 2P]$ of integers, coefficients $a_m \in \mathbb{C}$ and a height h_0,

$$W(\chi) = \sum_{m \in S} a_m \chi(m) m^{-ih_0}.$$

The sieve of Section 4 needs, for the integers with a factor m in a range $[x^{u_1}, x^{u_2}]$ and the rest a product of primes, that the non-trivial characters contribute less than the main term. In the Dirichlet polynomials above this is the following estimate. Write $\mathcal{X}^*$ for the non-trivial characters mod q.

Theorem 3.1 (the Type II estimate)**.** *Assume Inputs 1 and 2, and let $L = 11/5$, $u_1 = 43457/100000$ and $u_2 = 46959/100000$. Let $K_0 \geq 0$ be an integer, $e_1 > 0$ and $0 < \eta \leq 10^{-5}$. There is q_0, depending on f_0, K_0, e_1 and η, such that for every $q \geq q_0$, every integer P with $q^{u_1 L} \leq P \leq q^{u_2 L}$, every $S \subseteq (P, 2P]$, all a_m and h_0, every $K \leq K_0$, and all $Q_1, \dots, Q_K \geq q^{e_1}$ and $|h_1|, \dots, |h_K| \leq q^{\eta}$ with $Q_1 \cdots Q_K \leq 2^{K_0} q^L / P$,*

$$\sum_{\chi \in \mathcal{X}^*} |W(\chi)| \prod_{j=1}^{K} |S_{Q_j}(\chi, h_j)| \leq q^{L - 1/50000} \Big(\frac{1}{P} \sum_{m \in S} |a_m|^2 \Big)^{1/2}.$$

At $x = q^L$ the condition on P is $x^{u_1} \leq P \leq x^{u_2}$. By Cauchy–Schwarz and $Q_1 \cdots Q_K \leq 2^{K_0} q^L / P$, the trivial character's term is at most a constant times $q^L (\sum |a_m|^2 / P)^{1/2}$, and the bound saves the power $q^{-1/50000}$ against it. The proof has three parts: the prime sums are bounded through the zeros of each character's L-function, the characters are sorted by those zeros, and in each bin the sum over the characters is bounded in two ways. Sections 3.4 to 3.6 give the condition on the window under which this works, where it comes from, and its check at $L = 11/5$.

3.1 Prime sums in a zero-free rectangle

Lemma 3.2. *Assume Input 1. For every integer $J \geq 0$ there are $B, C \geq 0$ such that the following holds for every $q \geq 1$, every $\chi \in \mathcal{X}^*$, every $1/2 \leq \theta \leq 11/12$, $0 < \delta \leq 1/24$, $H \geq 0$, $Q \geq 4$ and $y \in \mathbb{R}$. If $L(s, \chi) \neq 0$ for $\operatorname{Re} s > \theta$, $|\operatorname{Im} s - y| < H + 1$, then*

$$|S_Q(\chi, y)| \leq M_\delta(\theta)\, \lambda\, B\, \frac{Q^{\theta+\delta} + Q^{11/12+\delta}(1+H)^{-J}}{\log Q} + C\big(\sqrt{2Q} + 1\big)\big(\log_2(2Q) + 1\big),$$

where $\lambda = \log(q+2) + \log(|y|+4) + 1$, and $M_\delta(\theta) = C_{\log}(\theta, \theta+\delta) + C_{\log}(11/12, 11/12+\delta)$ depends only on θ and δ; $C_{\log}$ is the constant of the bound for L'/L in §S1.

The argument is the classical one for primes in progressions [21, Chapter 5], with a smooth weight in place of a sharp cut-off: Perron's formula for $-L'/L$ against the Mellin transform of the weight, a bound for L'/L in zero-free discs by a lemma of Landau's type, and a contour that runs along $\operatorname{Re} s = \theta + \delta$ at the heights near y and along $\operatorname{Re} s = 11/12 + \delta$ elsewhere. The only use of Input 1 is to move the line of integration to $\operatorname{Re} s = 11/12 + \delta$ and to bound L'/L on that line and to its right. §S1 gives the proof.

3.2 The bins

Let $I = 500000$ and $\beta_i = 1/2 + 5i/(12I)$ for $0 \le i \le I$, so that $\beta_0 = 1/2$, $\beta_I = 11/12$ and the bins have width $1/1200000$. For a height bound $T \ge 2$ the *bin* of $\chi \in \mathcal{X}^*$ is the largest $i < I$ such that $L(s,\chi)$ has a zero with $\operatorname{Re} s > \beta_i$ and $|\operatorname{Im} s| \le T$, and 0 if there is none. If χ is in the bin i then:

- $L(s,\chi)$ has no zero with $\operatorname{Re} s > \beta_{i+1}$ and $|\operatorname{Im} s| \le T$: for $i+1 < I$ by the choice of i, and for $i+1 = I$ by Input 1. So Lemma 3.2 applies with $\theta = \beta_{i+1}$ to every height $|y| \le T - H - 1$.
- For $i \ge 1$, χ is counted by $N^*(\beta_i, T; q)$, so a density curve g bounds the number of characters in the bin by $C(qT)^{g(\beta_i)}(1+\log qT)^{10}$. For $i = 0$ the bound $\varphi(q) \le qT$ is the same, with $g(1/2) = 1$ and $\max(C,1)$ in place of C, which the bound for every i then uses.

The bins depend on T, and so on all the zeros up to the height T, but not on the heights h_j; the density curve is therefore used only for the zeros up to the height $T \le 3q^\eta$, a small power of q. The proof takes $\delta = 10^{-6}$, $H = q^\eta$ and $T = 2H + 1 \le 3q^\eta$, so that every height $|h_j| \le q^\eta$ has $|h_j| + H + 1 \le T$, and Lemma 3.2 with $J = \lceil 5(1+L)/(12\eta) \rceil$. As $Q_j \ge q^{e_1} \ge 4$ for large q and $Q_1 \cdots Q_K \le 2^{K_0} q^L/P$, each $Q_j \le q^{1+L}$ for large q, so that $Q_j^{11/12+\delta}(1+H)^{-J} \le Q_j^{1/2+\delta}$. The hypothesis $Q_j \ge q^{e_1}$ gives $\lambda/\log Q_j \ll 1/e_1$, and M_δ is bounded on the $I+1$ values β_i. So for large q, $|S_{Q_j}(\chi, h_j)| \le 2Q_j^{\beta_{i+1}+2\delta}$ in the bin i, and the product of the prime sums is at most $4^{K_0}(q^L/P)^{\beta_i+\varpi}$ with $\varpi = 1/1200000 + 2\delta < 2.9 \cdot 10^{-6}$.

3.3 Two bounds in a bin

Let $\mathcal{X}_i$ be the set of the characters in the bin i. Since the prime sums are bounded on $\mathcal{X}_i$, it remains to bound $\sum_{\chi \in \mathcal{X}_i} |W(\chi)|$. There are two bounds.

1. *Orthogonality.* For $S \subseteq (P, 2P]$ and every t,

$$\sum_{\chi} \Big| \sum_{m \in S} c_m \chi(m) m^{-it} \Big|^2 \le (P+q) \sum_{m \in S} |c_m|^2,$$

summed over all characters mod q. Indeed the left side is $\varphi(q)$ times the sum, over the classes mod q prime to q, of the squared moduli of the sums over the classes; each class has at most $P/q + 1$ elements in $(P, 2P]$, and Cauchy–Schwarz in each class and $\varphi(q) \le q$ give the bound. With Cauchy–Schwarz over $\mathcal{X}_i$,

$$\sum_{\chi \in \mathcal{X}_i} |W(\chi)| \le |\mathcal{X}_i|^{1/2} \Big((P+q) \sum_{m \in S} |a_m|^2 \Big)^{1/2}.$$

2. *The Halász–Montgomery inequality* [27, 28], in the form over the characters mod q at one height:

$$\Big(\sum_{\chi \in \mathcal{X}_i} |W(\chi)| \Big)^2 \le e \sum_{m \in S} |a_m|^2 \cdot \big(2P |\mathcal{X}_i| + |\mathcal{X}_i|^2 \mathcal{H} \big),$$

where $\mathcal{H}$ bounds $|k_{2P}(\psi)|$ for every non-trivial character ψ mod q, with $k_Y(\psi) = \sum_{n \ge 1} \psi(n) e^{-n/Y}$. Indeed, with $|c_\chi| \le 1$ and $c_\chi W(\chi) = |W(\chi)|$, $\sum_{\chi \in \mathcal{X}_i} |W(\chi)| = \sum_m a_m m^{-ih_0} \sum_{\chi \in \mathcal{X}_i} c_\chi \chi(m)$; Cauchy–Schwarz with the weight $e^{-m/2P}$, which is at least e^{-1} on $(P, 2P]$, and extending the second sum to all $m \ge 1$ give $(\sum_\chi |W(\chi)|)^2 \le e \sum_m |a_m|^2 \sum_{\chi, \chi'} c_\chi \overline{c_{\chi'}} k_{2P}(\chi \overline{\chi'})$. The diagonal terms are at most $\sum_{n \ge 1} e^{-n/2P} \le 2P$, and the others at most $\mathcal{H}$, since $\chi \overline{\chi'}$ is non-trivial for $\chi \ne \chi'$.

For the kernels, the formalisation proves the following bound from Input 1, and the window condition of Section 3.4 is written for it. Let $\delta' = \varepsilon' = 1/1000$. There is C such that for every q, every non-trivial character ψ mod q and every $Y \ge 1$,

$$|k_Y(\psi)| \le C Y^{1/12 - \delta'} (2q)^{5/12 + \delta' + 2\varepsilon'}.$$

Its proof (§S1) writes $k_Y(\psi) = (2\pi i)^{-1} \int_{(2)} L(s, \psi) \Gamma(s) Y^s \, ds$, moves the line to $\operatorname{Re} s = 1/12 - \delta'$, and bounds $L(s, \psi)$ there by the functional equation and the Lindelöf-type bound that Input 1 gives on $\operatorname{Re} s = 11/12 + \delta'$, as the Riemann hypothesis gives the Lindelöf hypothesis [20, §14.2]. At $Y = 2P$ it gives $\mathcal{H} = C(2P)^{1/12 - \delta'} (2q)^{5/12 + \delta' + 2\varepsilon'}$; for $P = q^\mu$ the exponent of q is the

$b(\mu)$ of Section 3.4, at most $1/2 + 2\varepsilon'$ for $\mu \le 1$, where the bound is used. Partial summation and the Pólya–Vinogradov inequality [19, Chapter 23] give a bound of the same size without Input 1: since the weight $e^{-n/Y}$ decreases from below 1 to 0,

$$|k_Y(\psi)| \le \max_N \Big| \sum_{n \le N} \psi(n) \Big| \ll_\varepsilon q^{1/2+\varepsilon}$$

for every non-trivial ψ mod q and every $Y > 0$, and the window at $11/5$ would hold with it too (Section 3.6).

Orthogonality is the better of the two bounds when the bin is large, and the Halász–Montgomery inequality when it is small, which is the case for β near $11/12$, on the top piece $[5/6, 11/12]$ of the curve. The proof takes the better of the two in each bin.

3.4 The window condition

Write $P = q^\mu$, and let $\beta = \beta_i$. Since $T \le 3q^\eta$ and $g \le 1$, the bin has $|\mathcal{X}_i| \le 3Cq^{g(\beta)+\eta}(1 + \log qT)^{10}$, and the product of the prime sums is at most $4^{K_0} q^{(L-\mu)(\beta+\varpi)}$. Up to constants, powers of $1 + \log qT$ and the factor q^η, the two bounds make the bin's contribution at most $q^E(\sum |a_m|^2/P)^{1/2}$, with the exponents

$$E_1 = (L-\mu)(\beta+\varpi) + \frac{g(\beta)}{2} + \frac{\mu}{2} + \frac{\max(1,\mu)}{2}$$

by orthogonality, and, by the Halász–Montgomery inequality with $(2P|\mathcal{X}_i| + |\mathcal{X}_i|^2\mathcal{H})^{1/2} \le (2P|\mathcal{X}_i|)^{1/2} + |\mathcal{X}_i|\mathcal{H}^{1/2}$, the larger of

$$E_2 = (L-\mu)(\beta+\varpi) + \frac{g(\beta)}{2} + \mu,$$
$$E_3 = (L-\mu)(\beta+\varpi) + \frac{g(\beta)}{2} + \frac{\mu}{2} + \frac{g(\beta)+b(\mu)}{2},$$

with $b(\mu) = 5/12 + \mu/12 + \delta'(1-\mu) + 2\varepsilon'$ the exponent of $\mathcal{H}$. For $\mu \ge 1$, $E_2 = E_1$, so the Halász–Montgomery inequality never helps there.

A window is written $[\mu_1, \mu_2]$, in exponents of q for the scale $P = q^\mu$ of the Type II variable, or, as in Theorem 3.1 and Sections 4 to 6, $[u_1, u_2]$, in exponents of $x = q^L$, with $\mu_1 = u_1 L$ and $\mu_2 = u_2 L$; Sections 3.5 and 3.6 use both.

Definition 3.3. A window $[\mu_1, \mu_2]$ is *admissible* for a curve g at L if for every $1/2 \le \beta \le 11/12$, every $\mu_1 \le \mu \le \mu_2$ and every $0 \le \varpi \le 4 \cdot 10^{-6}$, with $\delta' = \varepsilon' = 1/1000$, either $E_1 \le L - 1/25000$, or $\mu \le 1$ and $\max(E_2, E_3) \le L - 1/25000$.

So at each point one of the two bounds has an exponent at most $L - 1/25000$. Equivalently, the better of the two has the exponent

$$E(\beta,\mu,\varpi) = (L-\mu)(\beta+\varpi) + \frac{g(\beta)}{2} + \frac{\mu}{2} + \frac{1}{2}\max\bigl(\mu, \min(1, g(\beta)+b(\mu))\bigr),$$

and the window is admissible exactly when $E \le L - 1/25000$ at every point: the Halász–Montgomery inequality replaces the modulus q of the large sieve by $|\mathcal{X}_i|\mathcal{H}$ when that is smaller.

Lemma 3.4 (one bin)**.** *Let $g \le 1$ be a function and $[\mu_1,\mu_2]$ a window admissible for g at L. Given $C_0, C_d, C_K \ge 0$, there is C' such that the following holds. Let $q > 1$, $\mu_1 \le \mu \le \mu_2$ with $P = q^{\mu}$ an integer, $1 \le T \le 3q^{\eta}$ with $\eta \ge 0$, $1/2 \le \beta \le 11/12$ and $0 \le \varpi \le 4\cdot 10^{-6}$. Let $\mathcal{X}$ be a set of at most $C_d(qT)^{g(\beta)}(1+\log qT)^{10}$ characters on which $\prod_{j=1}^{K}|S_{Q_j}(\chi,h_j)| \le C_0(q^L/P)^{\beta+\varpi}$, for some K, Q_j and h_j, and let $\mathcal{H} \le C_K(2P)^{1/12-\delta'}(2q)^{5/12+\delta'+2\varepsilon'}$ bound $|k_{2P}(\psi)|$ for every non-trivial character ψ mod q. Then for every $S \subseteq (P,2P]$ and all a_m and h_0,*

$$\sum_{\chi\in\mathcal{X}} |W(\chi)| \prod_{j=1}^{K} |S_{Q_j}(\chi,h_j)| \le C'(1+\log qT)^{10} q^{L-1/25000+\eta}\Bigl(\frac{1}{P}\sum_{m\in S}|a_m|^2\Bigr)^{1/2}.$$

Proof. The left side is at most $C_0(q^L/P)^{\beta+\varpi}\sum_{\chi\in\mathcal{X}}|W(\chi)|$, and the two bounds of Section 3.3 bound the sum. At (β,μ,ϖ) the window condition gives $E_1 \le L - 1/25000$ or $\max(E_2,E_3) \le L - 1/25000$. In the first case, the orthogonality bound with $(P+q)^{1/2} \le \sqrt{2}\,q^{\max(1,\mu)/2}$ and $|\mathcal{X}|^{1/2} \le (3C_d)^{1/2} q^{(g(\beta)+\eta)/2}(1+\log qT)^5$ gives the claim with the exponent $E_1+\eta/2$. In the second, the Halász–Montgomery bound, split as above, gives two terms: $(2P|\mathcal{X}|)^{1/2}$ carries the exponent $E_2+\eta/2$, and $|\mathcal{X}|\mathcal{H}^{1/2}$ the exponent $E_3+\eta$, with $(1+\log qT)^{10}$. □

Proposition 3.5. *Assume Input 1. Let g be a density curve with $g \le 1$ and $g(1/2) = 1$, and let $[\mu_1,\mu_2]$ be admissible for g at L. Then the conclusion of Theorem 3.1 holds with $q^{\mu_1} \le P \le q^{\mu_2}$ in place of $q^{u_1 L} \le P \le q^{u_2 L}$, with L in place of $11/5$.*

Proof. By the bins of Section 3.2 and the kernel bound of Section 3.3, each bin i satisfies the hypotheses of Lemma 3.4 at $\beta = \beta_i$ and $\varpi = 1/1200000 + 2\delta$, with constants that do not depend on q, so each contributes at most $C'(1+\log qT)^{10} q^{L-1/25000+\eta}(\sum|a_m|^2/P)^{1/2}$. There are I bins, and for large q, $IC'(1+\log qT)^{10} \le q^{1/100000}$, so the total is at most $q^{L-1/25000+\eta+1/100000}(\sum|a_m|^2/P)^{1/2}$. Since $\eta \le 10^{-5}$, $L - 1/25000 + \eta + 1/100000 \le L - 1/50000$. □

3.5 Where the window comes from

Neglect the margin $1/25000$ and the allowance ϖ, and take a piece $g(\beta) = A(1-\beta)$ of a density curve. Then $E_1 \le L$ says that

$$\Big(L - \mu - \frac{A}{2}\Big)(1-\beta) \ge \frac{1-\mu}{2} \quad \text{for } \mu \le 1, \qquad \mu \le L - \frac{A}{2} \quad \text{for } \mu \ge 1.$$

At $\mu = 1$ both read $L \ge 1 + A/2$, whatever β. So, neglecting the margins, orthogonality gives a window around $P = q$ exactly when $L > 1 + A/2$ for the largest exponent A of the curve, and the window shrinks to the point $\mu = 1$ at $L = 1 + A/2$. The halving of A comes from Cauchy–Schwarz over a bin, which takes the bin's size to the power $1/2$.

The exponents of Section 2 are largest at $\sigma = 5/7$, where Ingham's $3/(2-\sigma)$ and Chen, Gupta and Li's $4/(1+\sigma)$ are both $7/3$. With $\beta = 5/7$ and $A = 7/3$ the two conditions give

$$u_1 \ge \frac{35}{9L} - \frac{4}{3}, \qquad u_2 \le 1 - \frac{7}{6L},$$

which are 0.43434 and 0.46970 at $L = 11/5$. They meet at $L = 13/6$. Below it the bounds of Section 3.3 give no window at all with these exponents, whatever the certificate, since the Halász–Montgomery inequality does not help at $\beta = 5/7$, where the bins are large: there $E_3 \le L$ needs $L \ge 7/3 + \mu/8$, even with $\mathcal{H}$ as small as $P^{1/2}$.

Three other points of the curve matter. The upper end is set at $\sigma = 4/7$ as well: below $4/7$ the curve is the trivial count $g = 1$, which is $A(1-\sigma)$ with $A = 1/(1-\sigma)$, and A tends to $7/3$ at $4/7$, which gives the same condition $u_2 \le 1 - 7/(6L)$. This is why the curve takes the trivial count only below $4/7$; with the margins its bound for u_2 is $3.9 \cdot 10^{-5}$ from binding. Near $\beta = 11/12$ the factor $1-\beta$ of the first condition is small, and with the curve's exponent $101/50$ there orthogonality alone would need μ close to 1, $u_1 \ge 0.4373$ at $11/12$; there the bins are small, and the Halász–Montgomery inequality takes over (Section 3.6). So the lower end is set at $5/7$ only with its help. At $\sigma = 7/9$, where $4/(1+\sigma)$ meets Huxley's $3/(3\sigma - 1)$ at $A = 9/4$, the first condition gives $u_1 \ge 27/(10L) - 4/5$, which binds only for $L > 107/48 \approx 2.229$. For the classical exponents, whose largest is $12/5$, at $\sigma = 3/4$, the threshold is $1 + 6/5 = 11/5$: without Input 2 the window shrinks to a point at $L = 11/5$, and the classical estimates give only about 2.24 (§S6). That this threshold is the exponent of Theorem 1.1 is a coincidence: with Input 2 the threshold is $13/6$, and in both cases the sieve's certificate needs between 0.03 and 0.04 above the threshold.

3.6 The window at 11/5

Proposition 3.6. *The window* $[u_1 L, u_2 L]$*, with* $u_1 = 43457/100000$ *and* $u_2 = 46959/100000$*, is admissible for* g^* *at* $L = 11/5$*.*

Proof. On each piece of g^*, $g^*(\beta) = A(1-\beta)$ with A constant, or $g^* = 1$, so each E_i is affine in β on the piece. Each E_i increases with ϖ, since $L - \mu > 0$. As a function of μ, E_1 is convex: it decreases for $\mu < 1$, with slope $1/2 - \beta - \varpi \le 0$, and increases for $\mu > 1$, with slope $1 - \beta - \varpi > 0$. On the top piece $[5/6, 11/12]$, E_2 increases with μ, so for $\mu \le 1$ it is at most $E_2(\beta, 1, \varpi) = E_1(\beta, 1, \varpi) \le E_1(\beta, \mu_2, \varpi)$, and E_3 decreases with μ, with slope $13/24 - \delta'/2 - \beta - \varpi < 0$. As a function of β on a piece $A(1-\beta)$, E_1 has slope $L - \mu - A/2$, which is positive for $\mu \le \mu_2$ since every exponent of g^* is at most $2.3334 < 2(L - \mu_2)$ (on $g^* = 1$ the slope is $L - \mu > 0$), and on the top piece E_3 has slope $L - \mu - A < 0$. Since $\mu_1 < 1 < \mu_2$, it therefore suffices that $E_1 \le L - 1/25000$ at the right end β_+ of every piece below $5/6$ for $\mu \in \{\mu_1, \mu_2\}$, and at $(11/12, \mu_2)$ on the top piece, and that $E_3 \le L - 1/25000$ at $(5/6, \mu_1)$, all at $\varpi = 4 \cdot 10^{-6}$. These are 134 inequalities between rational numbers, the *corners* of the window, and each holds, the least with a slack of $6.3 \cdot 10^{-6}$. They hold also with $A_k = 4/(1+\sigma_k) + 10^{-4}$ in place of the table's values, the least then with a slack of $1.6 \cdot 10^{-6}$; since each E_i increases with g, they hold for every A_k up to that limit. Solved for μ, with $\Theta = L - 1/25000$, they say that μ_1 is at least $\big(L(\beta_+ + \varpi) + g_+/2 + 1/2 - \Theta\big)/(\beta_+ + \varpi - 1/2)$ at the right end β_+ of every piece below $5/6$, and μ_2 at most $\big(\Theta - L(\beta_+ + \varpi) - g_+/2\big)/(1 - \beta_+ - \varpi)$ at these ends and at $11/12$, with g_+ the piece's value at β_+, $A(1-\beta_+)$ or 1, besides the top piece's condition on E_3. Both bounds are set at $\beta_+ = 1201/1680$, the right end of the first step of $[5/7, 131/168]$, and give the exact window $[0.43455660\ldots, 0.46961060\ldots] \cdot L$, which contains the proposition's. At $\varpi = 0$, with no margin and $\beta_+ = 5/7$, $A = 7/3$, they reduce to the closed forms of Section 3.5. The 134 inequalities are checked in exact rational arithmetic (§S5). □

Between the closed forms of Section 3.5 and the exact window of the proof the losses are these, in units of 10^{-4} of u (adding the effects in this order):

	lower end u_1	upper end u_2
closed form	0.4343434	0.4696970
the exponent 7/3 used up to 1201/1680	+0.98	0
the margin 1/25000 of Definition 3.3	+0.85	−0.64
the excess $2.3334 - 7/3$ of the first step	+0.20	−0.15
the allowance $\varpi \le 4 \cdot 10^{-6}$	+0.11	−0.07
exact window	$0.43455660\ldots$	$0.46961060\ldots$
window used	0.43457	0.46959
least slack of a corner, in the exponent of q	$6.3 \cdot 10^{-6}$	$1.29 \cdot 10^{-5}$

The allowance $\varpi \le 4 \cdot 10^{-6}$ covers the bins' width 1/1200000 and the 2δ of Section 3.2, rounded up. At the upper end the first step costs nothing, since for $\mu \ge 1$ the condition $\mu \le L - A/2$ does not involve β. The window used, found by a sampled scan, lies $1.3 \cdot 10^{-5}$ and $2.1 \cdot 10^{-5}$ inside the exact one. With u_1 and u_2 fixed, the corners hold for every $L \ge 2.1999832\ldots$, so the window leaves about $1.7 \cdot 10^{-5}$ of room in the exponent (§S6).

Proof of Theorem 3.1. By Proposition 2.2, g^* is a density curve under the two inputs, with $g^* \le 1$ and $g^*(1/2) = 1$. Apply Proposition 3.5, under Input 1, with the window of Proposition 3.6. □

The Halász–Montgomery branch is used only on the top piece of the curve, for $\mu \le 1$, where the bins are small, and it is needed there: at $(11/12, \mu_1)$, E_1 exceeds $L - 1/25000$ by $2.5 \cdot 10^{-3}$, and orthogonality alone would need $u_1 \ge 0.4373$. It is needed because of the curve's exponent near 11/12: with a constant exponent of at most 1.9595 on $[7/8, 11/12]$, as Huxley's $3/(3\sigma - 1)$ would give, orthogonality alone would hold at every corner, but the density estimates proved here have no exponent below 1.97 near 11/12 (§S6). The kernel bound enters only through the corner of E_3 at $(5/6, \mu_1)$, which holds with a slack of 0.099. So the corners hold with any constant kernel exponent up to 0.697 in place of $b(\mu)$, and the Pólya–Vinogradov bound $\mathcal{H} \ll_\varepsilon q^{1/2+\varepsilon}$ of Section 3.3 would do: the window does not need Input 1 for the kernel bound.

No bound for the kernels could move the ends. At the upper end $\mu_2 > 1$, and there the Halász–Montgomery inequality never helps. At the lower end the binding corner is at $\beta = 1201/1680$, where $g^*(\beta) + b(\mu_1) = 1.164 > 1$, so that $E_3 > E_1$ and the inequality loses to orthogonality. Every valid bound $\mathcal{H}$ is at least about $P^{1/2}$, since the mean square of $|k_{2P}(\psi)|$ over the non-trivial characters mod q is of order $P\varphi(q)/q$ when $P \le q^{1-\varepsilon}$, and $g^*(\beta) + \mu_1/2 = 1.143 > 1$ still. So the ends are set by orthogonality, that is, by the density exponent near 5/7 and the large sieve.

4 Type I estimates and Harman's sieve

Let $2 < L \le 5/2$ and $0 < u_1 < u_2 < 1$; this section uses nothing else of L and the window, and Section 5 adds the conditions of Theorem 6.1. Below, 'the conclusion of Theorem 3.1' means the Type II estimate of that theorem at this L on the window $[u_1 L, u_2 L]$, which Proposition 3.5 gives on every admissible window; Theorem 3.1 is its case $L = 11/5$, $u_1 = 43457/100000$, $u_2 = 46959/100000$. Let $x = q^L$ and let a be prime to q. The *window's range* is $[x^{u_1}, x^{u_2}]$, the range of the scale P of the Type II variable, which runs over $(P, 2P]$, and its *reflection* is $[x^{1-u_2}, x^{1-u_1}]$. The sets of the sieve are

$$\mathcal{A} = \{n \in (x, 2x] : n \equiv a \pmod q\}, \qquad \mathcal{B} = \{n \in (x, 2x] : (n, q) = 1\},$$

whose integers are those of $(N_1, N_2]$ with $N_1 = \lfloor x \rfloor$ and $N_2 = \lfloor 2x \rfloor$. For a set $\mathcal{C}$ of integers, a positive integer r and $z \ge 1$, $\mathcal{C}_r$ is the set of the multiples of r in $\mathcal{C}$, and $S(\mathcal{C}_r, z)$ the number of $n \in \mathcal{C}_r$ with no prime factor of n/r below z. When r has no prime factor below z, as in every use below, this is the number of multiples of r in $\mathcal{C}$ with no prime factor below z. With $z_{\mathrm{top}} = \lfloor N_2^{1/2} \rfloor + 1$, an integer of $(N_1, N_2]$ with no prime factor below z_{top} is prime, so $S(\mathcal{A}, z_{\mathrm{top}})$ is the number of primes in $\mathcal{A}$. The sieve compares $\mathcal{A}$ with $\mathcal{B}/\varphi(q)$ through the weight

$$\mathbf{1}[n \equiv a] - \frac{\mathbf{1}[(n, q) = 1]}{\varphi(q)} = \frac{1}{\varphi(q)} \sum_{\chi \in \mathcal{X}^*} \overline{\chi}(a)\chi(n),$$

which is orthogonality: the trivial character cancels, and the other characters carry the difference. This section states the tools of Harman's sieve [38], [39, Chapter 3] in the form the proof uses. Lemma 4.1 and Lemmas 4.4 to 4.6 use standard arguments, among them a Fourier separation of variables, which §S2 gives in full, and Proposition 4.7 is Harman's comparison. Lemma 4.2 is a bilinear form of Type I information, proved by orthogonality, Cauchy–Schwarz and Parseval's identity, with no bound for Kloosterman sums; Lemma 4.3 applies it at the leaves of Buchstab's tree, and Proposition 4.8 is Harman's comparison with it. The proofs of Lemmas 4.4 and 4.3 and of Propositions 4.7 and 4.8 are in the supplement (§S2 and §S3).

4.1 Type I

Lemma 4.1. *Let $0 \le N \le N'$ be integers and let $M > 0$ be prime to q. The number of $n \in (N, N']$ with $n \equiv a \pmod q$ and $M \mid n$ is $(N' - N)/(Mq)$*

within 1*, and the number of* $n \in (N, N']$ *prime to* q *with* $M \mid n$ *is* $\varphi(q)(N' - N)/(Mq)$ *within* $\varphi(q)$*.*

Proof. Since $(M, q) = 1$, the Chinese remainder theorem gives c with $c \equiv 0 \pmod{M}$ and $c \equiv a \pmod{q}$, and the two conditions $n \equiv a \pmod{q}$ and $M \mid n$ together say $n \equiv c \pmod{Mq}$. For $r \geq 1$ the number of $n \in (N, N']$ with $n \equiv c \pmod{r}$ is $\lfloor (N' - c)/r \rfloor - \lfloor (N - c)/r \rfloor$, which differs from $(N' - N)/r$ by less than 1. With $r = Mq$ this is the first count. The integers prime to q form the $\varphi(q)$ residue classes b mod q with $(b, q) = 1$, and adding the first count over these classes gives the second. □

So $|\mathcal{A}_M| - |\mathcal{B}_M|/\varphi(q)$ is at most 2 in absolute value for every M prime to q. It is used at the leaves of Proposition 4.7, where what matters is that the number of counts, at most a constant times q^{L-1-2c} for a fixed $c > 0$ (§S3), keeps their errors small against the size x/q of the main terms; the size of each modulus does not. So these counts serve moduli up to about x/q. Beyond x/q the sieve uses Type I information in a bilinear form, for moduli that are products $d_1 d_2$ of two parts of suitable sizes.

4.2 Type I beyond x/q

Write $e_q(t) = e^{2\pi i t/q}$ and $\vartheta(n) = \mathbf{1}[n \equiv a] - \mathbf{1}[(n, q) = 1]/\varphi(q)$, the weight of the comparison. For a weight w on a finite set S of integers, let

$$\hat{w}(h) = \sum_{k \in S} w(k)\, e_q(hk) \quad (h \bmod q),$$

$$E_w = \sum_{k \in S} w(k) \mathbf{1}[(k, q) = 1] - \frac{\varphi(q)}{q} \sum_{k \in S} w(k).$$

Lemma 4.2 (a bilinear Type I estimate)**.** *Let* $\mathcal{D}_1$ *and* $\mathcal{D}_2$ *be finite sets of positive integers prime to* q*, with coefficients* $\alpha(d_1)$ *and* $\beta(d_2)$*, and let* w *be a weight on a finite set* S *of integers. Let* $H \geq 0$ *and* $D \geq 1$ *be integers with* $\mathcal{D}_2 \subseteq [1, D]$ *and* $2HD < q$*, and suppose that every residue class mod* q *contains the inverses of at most* m *elements of* $\mathcal{D}_1$*, that every integer* $|t|d$ *with* $0 < |t| \leq H$ *and* $d \in \mathcal{D}_2$ *has at most* F *divisors, and that* $W \geq 0$ *and* $|\hat{w}(h)| \leq W$ *at every* $h \neq 0$ *that is not the residue of an integer* t *with*

$0 < |t| \le H$. *Then*

$$\Big|\sum_{d_1,d_2}\alpha(d_1)\beta(d_2)\sum_{k\in S}w(k)\,\vartheta(d_1d_2k)\Big| \le \Big(\sum|\alpha|^2\Big)^{1/2}\Big(mF\sum|\beta|^2\cdot\frac{1}{q}\sum_h|\hat w(h)|^2\Big)^{1/2} + W\sum|\alpha|\sum|\beta| + \frac{|E_w|}{\varphi(q)}\Big|\sum\alpha\sum\beta\Big|.$$

Proof. For d prime to q, with $\bar d$ its inverse mod q, $\mathbf{1}[dk\equiv a] = q^{-1}\sum_h e_q(h(k-a\bar d))$, and $(dk,q)=1$ exactly when $(k,q)=1$. The frequency $h=0$ contributes $q^{-1}\sum_k w(k)$, and with the second part of ϑ it leaves $-E_w/\varphi(q)$, so that

$$\sum_{d_1,d_2}\alpha(d_1)\beta(d_2)\sum_k w(k)\,\vartheta(d_1d_2k) = \frac{1}{q}\sum_{h\neq0}\hat w(h)B(h) - \frac{E_w}{\varphi(q)}\sum\alpha\sum\beta,$$
$$B(h) = \sum_{d_1,d_2}\alpha(d_1)\beta(d_2)\,e_q(-ha\bar d_1\bar d_2).$$

Since $2H<q$, the residues of the integers t with $0<|t|\le H$ are distinct. At the other frequencies $|\hat w(h)|\le W$ and $|B(h)|\le\sum|\alpha|\sum|\beta|$, and there are fewer than q of them, which gives the second term. At the frequencies t, Cauchy–Schwarz over d_1 gives

$$\Big|\sum_t\hat w(t)B(t)\Big|^2 \le \sum|\alpha|^2\cdot\sum_{d_1}\Big|\sum_{t,d_2}\hat w(t)\beta(d_2)\,e_q(-at\bar d_2\cdot\bar d_1)\Big|^2,$$

and as d_1 runs over $\mathcal{D}_1$, $\bar d_1$ runs over the residues mod q, each at most m times. Parseval's identity, $\sum_{s \bmod q}|\sum_j c_je_q(sf_j)|^2 = q\sum_{f_j\equiv f_{j'}}c_j\bar c_{j'}$, with $j=(t,d_2)$, $c_j=\hat w(t)\beta(d_2)$ and $f_j=-at\bar d_2$, bounds the inner sum by mq times the sum of $|\hat w(t)\beta(d_2)\hat w(t')\beta(d_2')|$ over the pairs with $t\bar d_2\equiv t'\bar d_2'$, that is $td_2'\equiv t'd_2 \pmod q$. Since $|td_2'|$ and $|t'd_2|$ are at most $HD<q/2$, the congruence is the equation $td_2'=t'd_2$. Bound each product by $\frac12(|\hat w(t)|^2|\beta(d_2')|^2+|\hat w(t')|^2|\beta(d_2)|^2)$; for given (t,d_2') the pairs (t',d_2) with $t'd_2=td_2'$ number at most the divisors of $|t|d_2'$, at most F. So the inner sum is at most $mqF\sum_t|\hat w(t)|^2\sum|\beta|^2$, and $\sum_t|\hat w(t)|^2\le\sum_h|\hat w(h)|^2$. Dividing by q gives the first term. □

No bound for Kloosterman sums enters: the inverses $\bar d_1$ are averaged by Parseval's identity, and the inverses $\bar d_2$ survive only on the diagonal $td_2' =$

$t'd_2$, which $2HD < q$ makes an equation in integers. The lemma is applied with w a smooth majorant or minorant of the cofactor's range, a spline: for an interval $(A, B]$, a step $\ell \geq 1$ and a power $r \geq 1$, $w(k)$ is ℓ^{-r} times the number of $(i, j_1, \dots, j_r)$ with $A < i \leq B$, $0 \leq j_s < \ell$ and $i + j_1 + \cdots + j_r = k$. Then $0 \leq w \leq 1$, $w = 1$ on $(A + r(\ell - 1), B]$, $w = 0$ off $(A, B + r(\ell - 1)]$, $\sum_h |\hat{w}(h)|^2 \leq q(\lfloor n/q \rfloor + 1)n$ with $n = B - A + r(\ell - 1)$, $|E_w| \leq d(q)$, and, since $\hat{w}(h)$ is the sum over the interval times the r-th power of a geometric sum of length ℓ, $|\hat{w}(h)| \leq (B - A)(q/(2\ell(H + 1)))^r$ at the residues h of the integers t with $H < |t| \leq q/2$ (§S3). In the application the coefficients are bounded, and the pairs (d_1, d_2) are taken in short blocks, on each of which $d_1 \approx P_1$, $d_2 \approx P_2$ and the cofactor runs over an interval of length about $n = x/(P_1P_2)$, with a spline whose step ℓ is below n by a small power of q. Measured against x/q, the size of a block's main term, the first term of the lemma is then about $(F(P_1 + q)(n + q)/x)^{1/2}$: Parseval's identity over the inverses $\bar{d}_1$ costs the factor $P_1/q + 1$, the cofactor's residues the factor $n/q + 1$, and the diagonal leaves F. At the leaves of Lemma 4.3 below, the cofactor is short, $n \leq 2q^{1+\kappa_0}$ when $d_1d_2 \geq x/q^{1+\kappa_0}$, and if $d_1 \leq x/q^{1+\kappa_1}$ with $\kappa_0 < \kappa_1 \leq L - 2$ the first term is at most about $(Fq^{\kappa_0 - \kappa_1})^{1/2}$ times the main term, whatever d_2: so d_1d_2 may exceed x/q, and only d_1, the part that Parseval's identity averages, must stay below it. The second term, with $D = P_2$ and $H = \lfloor (q - 1)/(2D) \rfloor$, is about $q(P_2/\ell)^r$ times the main term, which is small when d_2 is below the cofactor's length n by a power of q, that is when $d_1d_2^2$ is below x by a power of q, and r is large; and the third is small since then $d_1d_2 \leq xq^{-1/2}$. The two conditions allow moduli d_1d_2 up to about $x/q^{1/2}$, against x/q for Lemma 4.1.

The sieve meets these sums at the leaves of Buchstab's tree, the integers that are multiples of a product d of primes and have no other prime factor below a small power w of q. Sifting by the primes below w with Brun's weights and splitting the primes of d into two groups gives sums of the kind of Lemma 4.2, which yields the following lemma. Here and below, a *comparison* of tuples of primes is a condition $c + \sum_i e_i \log y_i > 0$ (or ≥ 0) on their entries, with a constant term c and coefficients $e_i \in \{-1, 0, 1\}$; an index i with $e_i = \pm 1$ is a *pivot* of it. The constant C_0 bounds the sums of $1/p$ over the ranges $[y, y^2)$, and $\zeta_k = e^{2C_0}(2C_0^k/k!) \exp(e^{2C_0} \cdot 2C_0^{k+1}/(k+1)!)$, which tends to 0 as the even level k grows (§S3).

Lemma 4.3 (the leaves by the bilinear estimate)**.** *Let $L > 2$, $0 \leq \kappa_0 < \kappa_1 \leq L - 2$, $0 < \kappa_2 \leq L$, $\varepsilon_w > 0$, $\Lambda_0 \geq 0$, integers K_1, K_2 and J_0, and an even $k \geq 2$ with $2C_0^2 \leq (k + 1)(k + 2)$ and $(2k + 4)\varepsilon_w \leq 1/16$. There are $\delta > 0$ and q_0 such that the following holds for every $q \geq q_0$ and every a prime to*

q, with $w = \lceil q^{\varepsilon w} \rceil$. Let $\mathbf{y}_1$ and $\mathbf{y}_2$ run over sets of strictly decreasing K_1- and K_2-tuples of primes in $[w, N_2]$, cut out by at most J_0 comparisons, each with a pivot and with its form at most $\Lambda_0 \log q$, and suppose that wherever the comparisons hold, the products $d_1 = \prod \mathbf{y}_1$ and $d_2 = \prod \mathbf{y}_2$ satisfy

$$d_1 w^{2k+4} \le q^{L-1-\kappa_1}, \qquad d_1 w^{2k+4} d_2^2 \le q^{L-\kappa_2}, \qquad q^{L-1-\kappa_0} \le d_1 d_2.$$

Then, with $d = d_1 d_2$,

$$\Big| \sum_{\mathbf{y}_1, \mathbf{y}_2} \Big(S(\mathcal{A}_d, w) - \frac{S(\mathcal{B}_d, w)}{\varphi(q)} \Big) \Big| \le \frac{\zeta_k(N_2 - N_1)}{\varphi(q) \log w} \sum_{\mathbf{y}_1, \mathbf{y}_2} \frac{1}{d} + \frac{q^{L-\delta}}{\varphi(q)}.$$

The proof is in §S3. The first term is the sieve's: Brun's weights lose the factor ζ_k of the main term, as at a leaf of Proposition 4.7, while the leaves themselves, unlike those, may have d beyond x/q. The factor w^{2k+4} bounds the moduli of Brun's weights, which join d_1.

4.3 Type II

Theorem 3.1 bounds Dirichlet polynomials. The sieve needs sums over $\mathcal{A}$ and $\mathcal{B}$ of integers $n = mp_1 \cdots p_K$ whose primes are cut out by comparisons. Fix a smooth function $f_0 \ge 0$ supported in $[1/2, 2]$, integers $K_0, J_0 \ge 0$ and reals $e_1 > 0$ and $E_0, \Lambda_0, A_0 \ge 0$. A *family* consists of

- an integer scale P with $x^{u_1} \le P \le x^{u_2}$, a set $S \subseteq (P, 2P]$ and coefficients $|a_m| \le A_0$;
- between 1 and K_0 primes $p_i \le 2Q_i$ with weights $f_0(p_i/Q_i)$, at scales $Q_i \ge q^{e_1}$ with $Q_1 \cdots Q_K \le 2^{K_0} x/P$;
- at most J_0 comparisons, conditions on (m, p) that hold when $\ell_j(m, p) > 0$ and fail when $\ell_j(m, p) < 0$, where $\ell_j(m, p) = \lambda_j(m) + \sum_i e_{ji} \log p_i$ with $|e_{ji}| \le E_0$ and $\ell_j \le \Lambda_0 \log q$ for every $m \in S$ and all primes $p_i \le 2Q_i$, each comparison with a *pivot*, an index i with $e_{ji} = \pm 1$.

Its sum over $\mathcal{A} - \mathcal{B}/\varphi(q)$ is the sum of $a_m \prod_i f_0(p_i/Q_i)$ times the weight above at $n = mp_1 \cdots p_K$, over the (m, p) that satisfy the comparisons. The integers n need not lie in $(x, 2x]$; where they must, two comparisons say so.

Lemma 4.4. *Assume the conclusion of Theorem 3.1. Given f_0, K_0, e_1, J_0, E_0, Λ_0 and A_0, there are $\delta > 0$ and q_0 such that for every $q \ge q_0$, every a prime to q and every family, its sum over $\mathcal{A} - \mathcal{B}/\varphi(q)$ is at most $A_0 q^{L-\delta}/\varphi(q)$ in absolute value.*

The proof is in §S2. By orthogonality the family's sum is a sum over the non-trivial characters. Replacing each comparison by a smooth cut-off of its form makes the family smooth; Fourier inversion then separates the variables, and Theorem 3.1 bounds the smooth family at each frequency of size at most a small power of q; the decay of the cut-offs' Fourier transforms handles the larger ones. The difference between the sharp and the smooth comparisons lies near the boundaries $\ell_j = 0$; the proof bounds it by families of the same kind, again through Theorem 3.1, and by a count along the pivot.

Lemma 4.5. *Assume the conclusion of Theorem 3.1. Given K_0, J_0, $e_1 > 0$, $E_0 \geq 0$ and $\Lambda_1 \geq 0$, there are $\delta > 0$ and q_0 such that the following holds for every $q \geq q_0$ and every a prime to q. Let m run over a set $S \subseteq (P_1, P_2]$ of integers, for integers P_1 and P_2 with $q^{u_1 L} \leq P_1$ and $P_2 \leq q^{u_2 L}$, with coefficients $|a_m| \leq 1$, and let $1 \leq K \leq K_0$ primes $p_1, \dots, p_K$ run over all the primes up to $2x$. Let $J \leq J_0$ comparisons, with forms $\ell_j = \lambda_j(m) + \sum_i e_{ji} \log p_i$, $|e_{ji}| \leq E_0$, $\lambda_j(m) \leq \Lambda_1 \log q$ for $m \in S$, and pivots, cut out the (m, p), and suppose that wherever they all hold, for $m \in S$ and any primes p_i, every $p_i \geq 2q^{e_1}$ and $mp_1 \cdots p_K \leq 2x$. Then the sum of $a_m(\mathbf{1}[n \equiv a] - \mathbf{1}[(n, q) = 1]/\varphi(q))$ at $n = mp_1 \cdots p_K$ over the (m, p) that satisfy the comparisons is at most $q^{L-\delta}/\varphi(q)$ in absolute value.*

Proof. Split $(P_1, P_2]$ into the dyadic blocks $(2^b P_1, 2^{b+1} P_1]$, and each prime variable by a smooth partition of unity: take for the f_0 of Lemma 4.4 the fixed function $f_0(y) = F(y) - F(y/2)$, with F smooth and nondecreasing, 0 on $(-\infty, 1/2]$ and 1 on $[1, \infty)$. It is supported in $[1/2, 2]$, takes values in $[0, 1]$, and $\sum_{t \geq 0} f_0(y/2^t) = 1$ for every $y \geq 1$; the primes are at most $2x < 2^M$, $M = \lfloor \log_2(2x) \rfloor + 1$, so only the scales 2^t with $t \leq M$ occur. This writes the sum as a sum of at most $(3 + 4 \log q)^{K_0+1}$ pieces. In each, m runs over a block, which lies in $(P, 2P]$ with P in the window's range, and the primes carry the weights $f_0(p_i/2^{t_i})$ at the scales $Q_i = 2^{t_i}$; on its box every form is at most $(\Lambda_1 + K_0 E_0(L + 3)) \log q$. A piece with some $2^{t_i} < q^{e_1}$ vanishes, since its weights force $p_i < 2q^{e_1}$; and so does a piece with $\prod_i 2^{t_i} > 2^{K_0+1} x/P$, since its weights force $m \prod_i p_i > 2x$. In a piece the primes may run over the range $p_i \leq 2Q_i$ of a family instead of the primes up to $2x$, since a term with some $p_i > 2x$ violates the comparisons, which force $m \prod_i p_i \leq 2x$. Every other piece is a family, with $K_0 + 1$ in place of K_0, $A_0 = 1$ and $\Lambda_0 = \Lambda_1 + K_0 E_0(L + 3)$, and Lemma 4.4 bounds it by $q^{L-\delta_1}/\varphi(q)$. For large q the number of pieces is at most $q^{\delta_1/2}$, and $\delta = \delta_1/2$ will do. □

The sieve meets these sums with the rest of n given only as a rough number. The next lemma writes such sums as families. Its comparisons are those defined before Lemma 4.3; Proposition 4.7 asks each of its comparisons to have a pivot, and this lemma does not.

Lemma 4.6 (chains of primes)**.** *Assume the conclusion of Theorem 3.1. Given $k \ge 1$, J_0, $e_0 > 0$ and $\Lambda_1 \ge 0$, there are $\delta > 0$ and q_0 such that the following holds for every $q \ge q_0$ and every a prime to q. Let $\mathbf{r} = (r_1, \dots, r_k)$ run over the k-tuples of primes up to N_2 that satisfy at most J_0 comparisons with constant terms at most $\Lambda_1 \log q$, and suppose that every tuple of primes that satisfies the comparisons has $r_i \ge q^{e_0}$ for every i, and that one of the following holds for every such tuple, the same one for all:*

(a) $r_k \le r_i$ for every i, and $W_0 < r_1 \cdots r_k \le W_1$, for integers W_0 and W_1 with $W_1 \le N_1$, $q^{u_1 L} \le \lfloor N_1/W_1 \rfloor$ and $\lfloor N_2/(W_0+1) \rfloor \le q^{u_2 L}$;

(b) $r_1 > \cdots > r_k$, and $P_1 < \prod_{i \in \mathcal{I}} r_i \le P_2$ for a fixed set $\mathcal{I}$ of indices, for integers P_1 and P_2 with $q^{u_1 L} \le P_1$ and $P_2 \le q^{u_2 L}$;

(c) $r_1 > \cdots > r_k$, and $W_0 < \prod_{i \in \mathcal{J}} r_i \le W_1$ for a fixed set $\mathcal{J}$ of indices, for integers W_0 and W_1 as in (a).

Then, with $r = r_1 \cdots r_k$,

$$\Big| \sum_{\mathbf{r}} \Big(S(\mathcal{A}_r, r_k) - \frac{S(\mathcal{B}_r, r_k)}{\varphi(q)} \Big) \Big| \le \frac{q^{L-\delta}}{\varphi(q)}.$$

Proof. An integer n counted for $\mathbf{r}$ is $n = rm \in (N_1, N_2]$ with m free of prime factors below r_k; since every $r_i \ge r_k$, this says that n itself has no prime factor below r_k.

(a) Here m, the rest of n, is the Type II variable. Since $W_0 < r \le W_1$, m lies in $(\lfloor N_1/W_1 \rfloor, \lfloor N_2/(W_0+1) \rfloor]$, inside the window's range. So the sum is one of the kind of Lemma 4.5, with $a_m = 1$ on that interval and the primes $r_1, \dots, r_k$. Its comparisons are the given ones, each pivoted on a prime with a nonzero coefficient (one with none is a constant condition); the roughness of m, $\log \mathrm{lpf}(m) - \log r_k \ge 0$, with $\mathrm{lpf}(m)$ the least prime factor of m and the pivot r_k; and $N_1 < mr \le N_2$, two comparisons with the forms $\log m + \sum_i \log r_i - \log N_1$ and its counterpart. Their m-parts are at most $(L+1)\log q$. Wherever they hold, every $r_i \ge q^{e_0} \ge 2q^{e_0/2}$ and $mr \le 2x$, so Lemma 4.5 applies, with $K_0 = k+1$, $J_0 + 3$ comparisons, $E_0 = 1$, $e_0/2$ in place of e_1 and $\Lambda_1 + L + 1$ in place of Λ_1.

(b) Here $m = \prod_{i\in\mathcal{I}} r_i$ is the variable, and the rest of n must be written in primes. The cofactor n/r has no prime factor below $r_k \ge q^{e_0}$ and is at most $2x \le q^{L+1}$, so it is the product of a unique non-increasing tuple $y_1 \ge \cdots \ge y_s \ge r_k$ of primes with $s \le s_1 = \lceil (L+1)/e_0 \rceil$. For each s, regroup the pairs $(\mathbf{r}, y)$ by m: since $\mathbf{r}$ is strictly decreasing, m is a squarefree integer of $(P_1, P_2]$ with $|\mathcal{I}|$ prime factors, which, read in decreasing order, are the r_i with $i \in \mathcal{I}$, so m determines them. The other variables are the primes r_i with $i \notin \mathcal{I}$ and $y_1, \dots, y_s$. The conditions are the given comparisons (which imply $r_1 > \cdots > r_k$), $y_b \le y_a$ for $a \le b$, $y_a \ge r_k$ for every a, and $N_1 < m \prod_{i\notin\mathcal{I}} r_i \prod_l y_l \le N_2$, each a comparison whose m-part reads the r_i with $i \in \mathcal{I}$ off m. Those that involve only these r_i are conditions on m, and give coefficients $a_m \in \{0,1\}$ on $(P_1, P_2]$; each of the others has a nonzero coefficient at one of the other variables, a prime r_i with $i \notin \mathcal{I}$ or a y_l, which is its pivot. There are at most $J_0 + s_1^2 + s_1 + 2$ of them, with m-parts at most $(\Lambda_1 + 2L + 2)\log q$, and wherever they hold every prime is at least q^{e_0} and the product is at most $2x$. If $\mathcal{I} = \{1, \dots, k\}$ and $s = 0$, no prime is left: then $n = m \le P_2 < N_1$ for large q, and that sum is empty. Lemma 4.5, with $K_0 = s_1 + k + 1$, $J_0 + s_1^2 + s_1 + 2$ comparisons, $E_0 = 1$, $e_0/2$ in place of e_1 and $\Lambda_1 + 2L + 2$ in place of Λ_1, bounds each of the other sums by $q^{L-\delta_1}/\varphi(q)$; there are at most $s_1 + 1$ of them, and $\delta = \delta_1/2$ will do for large q.

(c) Here the variable is $m = n/\prod_{i\in\mathcal{J}} r_i$, which lies in the window's range as in (a); it is the product of the primes r_i with $i \notin \mathcal{J}$ and of the rough cofactor n/r. These primes are read back from m: since $m \le 2x \le q^{L+1}$, it has at most $s_1 = \lceil (L+1)/e_0 \rceil$ prime factors at least q^{e_0}, and the r_i with $i \notin \mathcal{J}$ are those of given ranks among them. For each choice of the ranks (a *slice*, of which there are at most 2^{s_1}), the pairs $(\mathbf{r}, n)$ with these ranks correspond to the pairs of the tuple $(r_i)_{i\in\mathcal{J}}$ and an m in the window's range with that many prime factors at least q^{e_0}; the comparisons on the r_i with $i \notin \mathcal{J}$ and the roughness of n/r become comparisons of the r_i with $i \in \mathcal{J}$ against parts read off m, each with a pivot among those r_i, as in (b). So each slice is one sum of the kind of Lemma 4.5, and $\delta = \delta_1/2$ will do for large q. With $\mathcal{J} = \{1, \dots, k\}$ no prime is read off m, and the argument is that of (a). □

In (a) the product r, and in (c) the product of the r_i with $i \in \mathcal{J}$, lies in the reflection of the window, and in (b) a product of some of the primes lies in the window. If $W_0 + 1 \ge x^{1-u_2'}$ (for instance $W_0 = \lfloor x^{1-u_2'} \rfloor$) and $W_1 = \lfloor x^{1-u_1'} \rfloor$ with $u_1 < u_1' < u_2' < u_2$, the conditions of (a) and (c) hold for large q, since $\lfloor N_1/W_1 \rfloor \ge x^{u_1'} - 2 \ge x^{u_1}$ and $\lfloor N_2/(W_0+1) \rfloor \le 2x^{u_2'} \le x^{u_2}$. Section 5 uses (a) and (c) in this way, with $u_1' = U_1/D$ and $u_2' = U_2/D$: the

margins absorb the factor 2 of the upper end $2x$ and the rounding of N_1/W_1.

4.4 Harman's comparison

Proposition 4.7 (Harman's comparison, in the form used)**.** *Assume the conclusion of Theorem 3.1. Fix $j \ge 0$, J_0, $\Lambda_1 \ge 0$, $\varepsilon_H > 0$, and exponents $a_0 > 0$ and b_0 with $a_0 + b_0 < (1-u_1)L$ and $(1-u_2)L < b_0 < L-1$. There are $\delta > 0$ and q_0 such that the following holds for every $q \ge q_0$ and every a prime to q. Let $(r_1, \dots, r_j)$ run over the ordered j-tuples of primes up to $2x$ that satisfy at most J_0 comparisons, each with a pivot and a constant term at most $\Lambda_1 \log q$, which force every $r_i \ge z_0 = \lceil q^{a_0} \rceil$ and $r = r_1 \cdots r_j \le q^{b_0}$. Then*

$$\Big| \sum_{(r_1,\dots,r_j)} \Big(S(\mathcal{A}_r, z_0) - \frac{S(\mathcal{B}_r, z_0)}{\varphi(q)} \Big) \Big| \le \frac{\varepsilon_H \, x}{\varphi(q) \log q} + \frac{q^{L-\delta}}{\varphi(q)}.$$

For $j = 0$ the sum is the single term $S(\mathcal{A}, z_0) - S(\mathcal{B}, z_0)/\varphi(q)$.

The proof (§S3) iterates Buchstab's identity [45] from z_0 down to a small power w of q, stopping a branch as soon as its product exceeds $X' = \lfloor q^{b_0} \rfloor$. A stopped branch has its product in $(X', z_0 X']$, the reflection of the window, and its cofactor in the window's range, so Lemma 4.5 evaluates the stops. The leaves are sifted only by the primes below w, and Brun's sieve [44], in the form of the fundamental lemma of sieve theory [22], [21, Chapter 6], compares their $\mathcal{A}$- and $\mathcal{B}$-counts, with an error that the elementary bound $\prod_{p<w}(1-1/p) \le 1/\log w$, Mertens' theorem for the sum over the leaves and the choice of the sieve's levels make at most $\varepsilon_H x/(\varphi(q)\log q)$.

The condition $r \le q^{b_0}$ with $b_0 < L-1$ keeps the leaves below x/q, where Lemma 4.1 serves. With Lemma 4.3 the leaves may be larger, if their primes split into two groups of the sizes that the lemma asks, and the stops may land in the window as well as in its reflection.

Proposition 4.8 (Harman's comparison with the bilinear estimate)**.** *Assume the conclusion of Theorem 3.1. Fix $j \ge 1$, J_0, $\Lambda_1 \ge 0$, $\varepsilon_H > 0$, an exponent a_0 with $0 < a_0 < (u_2 - u_1)L$, exponents $l_i \le h_i$ for $1 \le i \le j$, sets $\mathcal{S}_0, \mathcal{S}_1 \subseteq \{1, \dots, j\}$, a side $\iota \in \{1, 2\}$ and a real t. With $\sigma^- = \sum_{i \in \mathcal{S}_0} l_i$ and $\sigma^+ = \sum_{i \in \mathcal{S}_0} h_i$, suppose that*

$$u_1 L - \sigma^- \le t, \qquad \sigma^+ < (1-u_1)L,$$
$$(1-u_2)L - \max(u_2 L, \sigma^-) \le t \quad \text{if } \sigma^+ > u_2 L,$$

and that $D_1 = \sum_{i \in \mathcal{S}_1} h_i + \mathbf{1}[\iota = 1]\, t$ and $D_2 = \sum_{i \notin \mathcal{S}_1} h_i + \mathbf{1}[\iota = 2]\, t$ satisfy

$$D_1 < L - 1, \qquad D_1 + 2D_2 < L.$$

There are $\delta > 0$ and q_0 such that the following holds for every $q \geq q_0$ and every a prime to q. Let $(r_1, \dots, r_j)$ run over the j-tuples of primes up to $2x$ that satisfy at most J_0 comparisons, with constant terms at most $\Lambda_1 \log q$, which force $r_1 > \cdots > r_j \geq z_0 = \lceil q^{a_0} \rceil$ and $q^{l_i} \leq r_i \leq q^{h_i}$ for every i. Then, with $r = r_1 \cdots r_j$,

$$\Big| \sum_{(r_1,\dots,r_j)} \Big(S(\mathcal{A}_r, z_0) - \frac{S(\mathcal{B}_r, z_0)}{\varphi(q)} \Big) \Big| \leq \frac{\varepsilon_H \, x}{\varphi(q) \log q} + \frac{q^{L-\delta}}{\varphi(q)}.$$

Here $r_{\mathcal{S}_0} = \prod_{i \in \mathcal{S}_0} r_i$ is the part of r that the primes below z_0 bring into the window or its reflection, and t bounds the exponent of the factor they must supply: at most $u_1 L - \sigma^-$ from below the window, and at most $(1-u_2)L - \max(u_2 L, \sigma^-)$ from between the window and its reflection; $r_{\mathcal{S}_0}$ is never past the reflection. The proof (§S3) splits the sum by the position of $r_{\mathcal{S}_0}$. Where it is in the window, or in its reflection, one step of Buchstab's identity and Lemma 4.6, (b) or (c), evaluate it. Where it is below the window, or between the window and its reflection, Buchstab's tree from z_0 down to w stops a branch as soon as $r_{\mathcal{S}_0}$ times its primes reaches the window, or the reflection, and Lemma 4.6 evaluates the stops; so the primes P of a leaf have $\prod P < 4q^t$. A leaf with $d = r \prod P$ at most $q^{L-1-4c'}$, for a small $c' > 0$ (§S3), is a Type I count, as in Proposition 4.7; the others are split into d_1, the primes of $\mathcal{S}_1$ and, if $\iota = 1$, those of P, and d_2, the rest, and $D_1 < L - 1$ and $D_1 + 2D_2 < L$ are the conditions of Lemma 4.3, with margins.

5 The decomposition and its certificate

This section bounds $S(\mathcal{A}, z_{\mathrm{top}})$ from below. The decomposition writes it by Buchstab's identity as a signed sum of sifted terms. Each term is evaluated, that is compared with its $\mathcal{B}$-version by Lemma 4.6 or Propositions 4.7 and 4.8, or, if it has sign +, dropped, which costs its $\mathcal{B}$-version. The certificate is an upper bound for the total cost, computed exactly in integers. Throughout the section, L, the window and the grid satisfy the conditions of Theorem 6.1 (Section 6).

5.1 Exponents and the grid

The exponent of a prime p is $\alpha = \log p / \log x$. Fix a denominator $D = 2^{40}$ and integers $U_1 < U_2$ with $u_1 D < U_1$ and $U_2 < u_2 D$, so that $u_1 < U_1/D < U_2/D < u_2$; the grid of Proposition 5.4 rounds the window's ends inward, $U_1 = \lceil u_1 D \rceil$ and $U_2 = \lfloor u_2 D \rfloor$, which qualify since $u_1 D$ and $u_2 D$ are not

integers. Let $v = (U_2 - U_1)/D$. The decomposition sifts the primes of exponent at least v. The grid has N cells a side, of side $h = \lceil (D/2 - (U_2 - U_1))/N \rceil / D$; the cell k is the range $[e_k, e_k + h)$ of exponents with $e_k = v + kh$, for $0 \le k < N$, so that the cells cover $[v, v + Nh)$, and $v + Nh \ge 1/2$ (Theorem 6.1 asks $v + Nh > 1/2$, which holds on the grid of Proposition 5.4). Here $N = 272$. The primes of the four deepest levels of the decomposition, p_5 to p_8, have their exponents in the cells of a *coarse grid* with the same window and $N_C = N/4 = 68$ cells a side, each four consecutive cells of the grid: its side is $h_C = 4h$, and its cell b is $[e_b^C, e_b^C + h_C)$ with $e_b^C = v + bh_C = e_{4b}$.

The set of *evaluable* exponents is

$$\mathcal{U} = (U_1/D,\ U_2/D) \cup (1 - U_2/D,\ 1 - U_1/D).$$

For a box, a product of cells $c_1, \ldots, c_k$, with lower ends e_{c_i} ($e_{c_i}^C$ for a coarse cell) and widths w_i (h, or h_C for a coarse cell), and a set $\mathcal{S}$ of its coordinates, write $s^-(\mathcal{S}) = \sum_{i \in \mathcal{S}} e_{c_i}$ and $s^+(\mathcal{S}) = \sum_{i \in \mathcal{S}} (e_{c_i} + w_i)$; s^+ of all the coordinates is the box's *upper sum*. The box is *surely evaluable* if $[s^-(\mathcal{S}), s^+(\mathcal{S})] \subset \mathcal{U}$ for some set $\mathcal{S}$ of its coordinates, so that the sum of those exponents lies in $\mathcal{U}$ throughout the box. The ends are rational, so a box is tested exactly.

The parameters. Apply Proposition 4.7 with $a_0 = vL$ and $b_0 = (1 - U_2/D)L$, so that $z_0 = \lceil x^v \rceil$ and $X' = \lfloor x^{1-U_2/D} \rfloor$, and Proposition 4.8 with the same a_0. The hypotheses of the first hold: $a_0 + b_0 = (1 - U_1/D)L < (1 - u_1)L$, since $U_1/D > u_1$; $b_0 > (1 - u_2)L$, since $U_2/D < u_2$; and $b_0 < L - 1$ is the condition $LU_2 > D$ of Theorem 6.1. For large x, $z_{\text{top}} \le X'$, since $2U_2 < D$, and $z_0 X' \le x^{1-U_1/D}$ up to rounding: v is the largest floor for which the stops of Proposition 4.7 stay in the reflection of the rounded window $[U_1/D, U_2/D]$.

5.2 The rule

A term of the decomposition is decomposed only if its box passes the following test, the *rule*, which makes the term $S(\mathcal{A}_{p_1 \cdots p_k}, z_0)$ that its decomposition creates one that Proposition 4.7 or Proposition 4.8 evaluates. A box passes the rule if

(R1) its upper sum is below $1 - U_1/D$; or

(R2) it passes the *bilinear test* and has *no boundary term*, that is, its upper sum plus the upper end of its last cell is below 1.

For a set $\mathcal{S}$ of the box's coordinates with $s^+(\mathcal{S}) < 1 - U_1/D$, the empty set included, let

$$t(\mathcal{S}) = \max\Big(\frac{U_1}{D} - s^-(\mathcal{S}),\ \ \mathbf{1}\Big[s^+(\mathcal{S}) \ge \frac{U_2}{D}\Big]\Big(1 - \frac{U_2}{D} - \max\Big(s^-(\mathcal{S}), \frac{U_2}{D}\Big)\Big),\ \ 0\Big),$$

the largest exponent of a product of smaller primes with which the product of the primes of $\mathcal{S}$ may still enter the window from at or below it, or its reflection from between the two; let T be the least $t(\mathcal{S})$ over these sets, so that $T \le t(\emptyset) = U_1/D$. The box passes the *bilinear test* if its coordinates split into two sets $\mathcal{S}_1$ and $\mathcal{S}_2$, with T joined to one of them, such that

$$a_1 < 1 - \frac{1}{L}, \qquad a_1 + 2a_2 < 1,$$

where a_ι is $s^+(\mathcal{S}_\iota)$, plus T if T is joined to $\mathcal{S}_\iota$. A box that passes (R1) is *split* when it is decomposed, and a box that passes (R2) but not (R1) is taken *whole*.

(R1) is the test of a split: on a box that passes it, $\alpha_1 + \cdots + \alpha_k < 1 - U_1/D$, so that the part of the term with $p_1 \cdots p_k > X'$ has $X' < p_1 \cdots p_k < x^{1-U_1/D}$, and Lemma 4.6(a) applies to it, with $W_0 = X'$ and $W_1 = \lfloor x^{1-U_1/D} \rfloor$: the cofactor lies in $(\lfloor N_1/W_1 \rfloor, \lfloor N_2/(X'+1) \rfloor]$, which is inside the window's range for large q by the remark after Lemma 4.6. (R2) is the test of Proposition 4.8, in exponents of x: with l_i and h_i the ends of the cells times L, $\mathcal{S}_0$ a set at which T is attained, $t = TL$, and the split $(\mathcal{S}_1, \mathcal{S}_2)$ with ι the side of T, its hypotheses hold, since $U_1/D > u_1$ and $U_2/D < u_2$, and $a_0 = vL < (u_2 - u_1)L$. The condition of no boundary term makes empty the terms of depth $k+1$ below a whole box that count only products of $k+1$ primes (*Empty terms*, below); below a split box they are empty anyway, since its decomposed part has $p_1 \cdots p_k \le X'$.

5.3 The decomposition

The tree. Buchstab's identity

$$S(\mathcal{A}_m, z) = S(\mathcal{A}_m, z_0) - \sum_{z_0 \le p < z} S(\mathcal{A}_{mp}, p),$$

applied to $S(\mathcal{A}, z_{\mathrm{top}})$ and repeated, introduces primes $p_1 > p_2 > \cdots$, each at least z_0, with exponents $\alpha_1, \alpha_2, \ldots$. A *term of depth k* is a sum of $S(\mathcal{A}_{p_1 \cdots p_k}, p_k)$ over the primes whose exponents lie in a box of cells, fine for $\alpha_1, \ldots, \alpha_4$ and coarse for $\alpha_5, \ldots, \alpha_8$; its sign is $(-1)^k$. To *decompose* a term is to apply the identity to it, to its part with $p_1 \cdots p_k \le X'$ if its box

is split and to all of it if its box is whole; this gives the term $S(\mathcal{A}_{p_1\cdots p_k}, z_0)$, which Proposition 4.7 evaluates for a split box and Proposition 4.8 for a whole one, with the cells, the order of the primes and the earlier splits as its comparisons, and terms of depth $k+1$. A term of depth k is treated by the first of the following rules that applies.

(E) If its depth is at most six and its box is surely evaluable, it is evaluated by Lemma 4.6.

(O) If its box lies in $\alpha_k \geq \alpha_{k-1}$, that is, the cell of α_k starts at or above the end of the cell of α_{k-1}, or if the lower corner $(e_{c_1}, \dots, e_{c_k})$ of its box, with c_i the cell of α_i (and $e^C_{c_i}$ in place of e_{c_i} for a coarse cell), has $e_{c_1} + \cdots + e_{c_{k-1}} + 2e_{c_k} > 1$, so that $\alpha_1 + \cdots + \alpha_{k-1} + 2\alpha_k \geq 1 + 1/D$ on the box, it is empty for $k \geq 3$ (below), and for $k = 2$ it is empty if its box lies in $\alpha_2 \geq \alpha_1$ and otherwise counts only products p_1p_2; these non-empty cells (i, j) of depth two, with $e_i + 2e_j > 1$, are the *corner* cells, and they are dropped, at a negligible cost (Lemma 5.2).

(S) Otherwise, a term with sign $-$ is decomposed. The part of a split term with $p_1 \cdots p_k > X'$ is evaluated by Lemma 4.6(a), with the cofactor as the Type II variable.

(D) A term with sign $+$ is dropped, or, if the certificate chooses it, decomposed as in (S). It may be chosen only if the rule passes its box and the boxes of its terms of depth $k+1$ to which (E) and (O) do not apply.

By (D), the box of every term with sign $-$ of depth at least three passes the rule; at depth one there is no split, since $p_1 < z_{\text{top}} \leq X'$: a term whose cell lies inside the window is evaluated by Lemma 4.6(b) with $\mathcal{I} = \{1\}$, and every other one is decomposed. For $k \geq 3$ the order of (E) and (O) does not matter, since the terms of (O) are empty. The terms of depth eight are empty or dropped, and not decomposed. At depth two a cell to which (E) and (O) do not apply is a *candidate*; at depths four and six such a cell is *counted*, and at depth eight a cell to which (O) does not apply. Table 1 sums up, and Figure 1 shows the cells of depth two.

Empty terms. Let a term of depth $k \geq 3$ lie in (O), below a decomposed term of depth $k-1$. If its box lies in $\alpha_k \geq \alpha_{k-1}$, it is empty, since $p_k < p_{k-1}$. Otherwise $\alpha_1 + \cdots + \alpha_{k-1} + 2\alpha_k \geq 1 + 1/D$ on its box, so for large x, $p_1 \cdots p_k^2 \geq x^{1+1/D} > 2x$, and a cofactor m with no prime factor below p_k is 1: the term counts only $n = p_1 \cdots p_k$. If the term above was split, $p_1 \cdots p_{k-1} \leq X'$ on it, and $p_k < p_{k-1} < X'^{1/2}$, since $p_{k-1}^2 < p_1p_2 \leq X'$; so $n < X'^{3/2} < N_1$, since $\frac{3}{2}(1 - U_2/D) < \frac{3}{2}(1 - 1/L) \leq 9/10$ by $LU_2 > D$ and $L \leq 5/2$. If it was

Depth	Sign	Grid	Terms to which (E) and (O) do not apply
1	−	fine	decomposed (no part above X')
2	+	fine	candidate cells: dropped, or decomposed
3	−	fine	decomposed
4	+	fine	counted cells: dropped, or decomposed
5	−	coarse	decomposed
6	+	coarse	counted cells: dropped, or decomposed
7	−	coarse	decomposed ((E) not tested)
8	+	coarse	counted cells: dropped ((E) not tested)

Table 1: The decomposition. A term of depth at most six over a surely evaluable box is evaluated (E); one whose box lies past the region $\alpha_k < \alpha_{k-1}$, $\alpha_1 + \cdots + \alpha_{k-1} + 2\alpha_k \le 1$ in the sense of (O) is empty, or at depth two counts only products of two primes. A decomposed box is split at X' if it passes (R1), and taken whole if it passes only (R2).

whole, its box has no boundary term, so $n < p_1 \cdots p_{k-1}\, p_{k-1} \le x^{1-1/D} < N_1$ for large x. Either way the term counts no integer of $(N_1, N_2]$. At depth two the same holds for $\alpha_2 \ge \alpha_1$, while outside $\alpha_1 + 2\alpha_2 \le 1$ the cofactor is 1 and the term counts the products $n = p_1 p_2 \in (x, 2x]$.

Evaluation. Let the box of a term of depth k be surely evaluable, with $\sum_{i\in\mathcal{I}} \alpha_i \in \mathcal{U}$ on it. If the sum lies in $(U_1/D, U_2/D)$, Lemma 4.6(b) applies with this $\mathcal{I}$, the cells, the order of the primes and the earlier splits as comparisons. If it lies in $(1 - U_2/D, 1 - U_1/D)$, Lemma 4.6(c) applies with $\mathcal{J} = \mathcal{I}$, and with W_0 and W_1 as in the remark after the lemma. Below a split box only the set of all the primes can have its sum in the reflection, since a proper subset's product is at most $p_1 \cdots p_{k-1} \le X' \le x^{1-U_2/D}$; then (c) is case (a), with the cofactor as the variable.

5.4 The cost of a dropped term

The cost of a dropped term is the number of its integers in $\mathcal{B}$, and their cofactors are rough numbers. Let $\Phi(y, z)$ be the number of integers in $(y, 2y]$ with no prime factor below z, and let

$$\omega^*(u) = \begin{cases} 0, & u < 1, \\ 1/u, & 1 \le u < 2, \\ (1 + \log(u-1))/u, & 2 \le u \le 3, \\ 0.565073, & u > 3. \end{cases}$$

On $[1,3]$ this is Buchstab's function ω, whose maximum on $[2,3]$ is $0.567143\ldots$, the root t of $te^t = 1$, at $u = 1+1/t$, where $(u-1)\log(u-1) = 1$ [46]; above 3 it is the constant that the induction of the next lemma needs.

Lemma 5.1. *For every U and $\varepsilon > 0$, for all large integers z and all $y \geq 1$ with $u = \log 2y/\log z \leq U$, $\Phi(y,z) \leq (\omega^*(u) + \varepsilon)\, y/\log z$.*

This is an upper bound of the kind that Buchstab's asymptotic for rough numbers gives [45], [23, Chapter III.6], with an explicit constant above 3. The proof (§S4) is an induction on u with Buchstab's inequality, the prime number theorem and a delay inequality for ω^* (§S4(d)). Above 3 that inequality holds exactly when the constant c_* of ω^* there is at least $(1+\log 2+0.5671433)/4 = 0.5650726\ldots$, and 0.565073 is that bound rounded up.

A dropped term of depth two over the cell (i,j) counts the $n = p_1p_2m \in (x,2x]$ whose cofactor m has no prime factor below p_2. By Lemma 5.1 and the prime number theorem its $\mathcal{B}$-version is, for large q, at most $x/\log x$ times

$$\iint_{\text{cell}} \omega^*\Big(\frac{1-\alpha_1-\alpha_2}{\alpha_2}\Big)\frac{d\alpha_1\, d\alpha_2}{\alpha_1\alpha_2^2},$$

plus ε, and similarly at depth $k \in \{4,6,8\}$ with $d\alpha_1\cdots d\alpha_k/(\alpha_1\cdots\alpha_{k-1}\alpha_k^2)$ and the argument $(1-\alpha_1-\cdots-\alpha_k)/\alpha_k$. The certificate bounds each integral by a bound Ω for ω^* on the cell's range of arguments times the integral of the rest, a product of integrals of $d\alpha/\alpha$ and one of $d\alpha/\alpha^2$ over the cells. It bounds the first kind from above by the trapezoid rule, since $1/\alpha$ is convex, and takes the second exactly:

$$\int_e^{e+h}\frac{d\alpha}{\alpha} = \log\Big(1+\frac{h}{e}\Big) \leq t(e,h) = \frac{h(2e+h)}{2e(e+h)},$$
$$\int_e^{e+h}\frac{d\alpha}{\alpha^2} = s(e,h) = \frac{h}{e(e+h)}.$$

It works in units of $1/\mathcal{K}$ with $\mathcal{K} = 2^{60}$: Ω is rounded up to a multiple of $1/\mathcal{K}$, and below a cell of depth two the factors t and s of each prime are rounded up to multiples of 2^{-64}. The bound Ω for ω^* on a cell of depth two, whose arguments lie in $[u_{\min}, u_{\max})$ with $u_{\min} = (1-e_i-e_j-2h)/(e_j+h)$ and $u_{\max} = (1+1/D-e_i-e_j)/e_j$, is found by cases, since ω^* is not monotone:

- 0 if $u_{\max} \leq 1$;
- 0.565073, rounded up in units of $1/\mathcal{K}$, if $u_{\min} \geq 3$;

- $1/\max(u_{\min}, 1)$, rounded up, if $u_{\max} \le 2$;
- otherwise the largest of $1/\max(u_{\min}, 1)$, if $u_{\min} < 2$; of a bound for ω^* on the part of the range in $[2, 3]$; and of 0.565073, if $u_{\max} > 3$.

On $[2, 3]$, $\omega^* = \omega$ increases up to 2.76 and decreases from 2.77 on (§S4), so the bound there is $\omega(u_{\max})$ if $u_{\max} \le 2.76$, $\omega(u_{\min})$ if $u_{\min} \ge 2.77$, and 0.5671433, at least its maximum $0.567143\ldots$ there, otherwise. At a rational $u \in [2, 3]$ the certificate bounds $\omega(u) = (1 + \log(u-1))/u$ by $\log(u-1) \le \ell_k + (y - 1/y)/2$, where $r_k = 1 + k/16 \le u - 1 < r_k + 1/16$, $y = (u-1)/r_k \ge 1$, and $\ell_k \ge \log r_k$ is one of seventeen rationals of a table, each checked against the exponential's series; the second term is $\log y \le (y - 1/y)/2$, the trapezoid rule again, which here errs by less than $4 \cdot 10^{-5}$. At depths four, six and eight Ω is found in the same way: for a box of cells $c_1, \ldots, c_k$ the arguments lie in $[u_{\min}, u_{\max})$ with

$$u_{\min} = \frac{1 - E - (w_1 + \cdots + w_k)}{e_{c_k} + w_k}, \qquad u_{\max} = \frac{1 + 1/D - E}{e_{c_k}},$$

where $E = e_{c_1} + \cdots + e_{c_k}$; at depth two these are the ends above. The *bound* of a cell is then

$$\Delta_2(i, j) = \lceil \mathcal{K}\Omega\, t(e_i, h)\, s(e_j, h) \rceil,$$
$$\Delta_k(c_1, \ldots, c_k) = \mathcal{K}\Omega\, \tilde{t}(c_1) \cdots \tilde{t}(c_{k-1})\, \tilde{s}(c_k) \quad (k = 4, 6, 8),$$

with $\mathcal{K}\Omega$ the rounded value and $\tilde{t}(c)$ and $\tilde{s}(c)$ the values of t and s at the lower end and the width of the cell c, rounded up to multiples of 2^{-64}.

Lemma 5.2. *For every $\varepsilon_c > 0$ and all large q, with $X = x/\log x$:*

1. *the $\mathcal{B}$-version of the dropped term of a candidate cell (i, j) is at most $(\Delta_2(i, j)/\mathcal{K} + \varepsilon_c)X$;*
2. *that of a dropped counted cell of depth $k \in \{4, 6, 8\}$ is at most $(\Delta_k/\mathcal{K} + \varepsilon_c)X$;*
3. *the corner cells together contribute $o(X)$.*

Proof. For a cell k, write $\mathcal{P}_k$ for the set of the primes p with $x^{e_k} \le p < x^{e_k + h}$.

(1) The $\mathcal{B}$-version of the term counts the $n \in (x, 2x]$ prime to q of the form $n = p_1 p_2 m$ with (p_1, p_2) in the cell and m without prime factors below p_2. Dropping the conditions $(n, q) = 1$, $p_2 < p_1$, $z_0 \le p_2$ and $p_1 < z_{\text{top}}$, and mapping n to m, it is at most

$$\sum_{p_1 \in \mathcal{P}_i} \sum_{p_2 \in \mathcal{P}_j} \Phi\Big(\frac{x}{p_1 p_2}, p_2\Big).$$

A candidate has $e_i + 2e_j \le 1$, and $e_j \ge v \ge 4h$, so $e_i + e_j + 2h < 1$ and $y = x/(p_1p_2) \ge 1$. Lemma 5.1, with $U = 2/e_j$, applies to $\Phi(y, p_2)$, with $u = \log 2y/\log p_2 = (1-\alpha_1-\alpha_2+\tau)/\alpha_2$ and $\tau = \log 2/\log x$. Since $\tau < 1/D$ once $\log x > D \log 2$, $u < u_{\max}$, and $u \ge u_{\min}$ since $\tau \ge 0$. So Ω also bounds $\omega^*(u)$, and $\Phi(y, p_2) \le (\Omega + \varepsilon_1)x/(p_1p_2 \log p_2)$ for large x. The term is at most

$$(\Omega + \varepsilon_1)\, x \sum_{p_1 \in \mathcal{P}_i} \frac{1}{p_1} \sum_{p_2 \in \mathcal{P}_j} \frac{1}{p_2 \log p_2}.$$

By the prime number theorem and partial summation, for large x,

$$\sum_{p \in \mathcal{P}_i} \frac{1}{p} \le \log\Big(1 + \frac{h}{e_i}\Big) + \varepsilon_1 \le t(e_i, h) + \varepsilon_1,$$

$$\log x \sum_{p \in \mathcal{P}_j} \frac{1}{p \log p} \le \frac{1}{e_j} - \frac{1}{e_j + h} + \varepsilon_1 = s(e_j, h) + \varepsilon_1.$$

These are the integrals of $d\alpha/\alpha$ and $d\alpha/\alpha^2$ over the cells; the first bound, $\log(1 + h/e_i) \le t(e_i, h)$, is $\xi \le \sinh \xi$ at $\xi = \log(1 + h/e_i)$, since $t(e, h) = \sinh \log(1 + h/e)$. Since $\Delta_2(i, j)/\mathcal{K} \ge \Omega\, t(e_i, h)\, s(e_j, h)$, the term is at most $(\Delta_2(i, j)/\mathcal{K} + \varepsilon_c)X$ once ε_1 is small.

(2) The same argument bounds a dropped counted cell of depth k, with $\Phi(x/(p_1 \cdots p_k), p_k)$, $k - 1$ sums $\sum 1/p$ and one sum $\log x \sum 1/(p \log p)$, the primes from p_5 on in coarse cells, and the rounded factors $\tilde{t} \ge t$ and $\tilde{s} \ge s$. Here $y \ge 1$ follows, as in (1), from the test of (O) at the lower corner of a counted cell, with $4h \le v$ at depth four, $4h + 2h_C \le v$ at depth six and $4h + 4h_C \le v$ at depth eight.

(3) A corner cell (i, j) has $e_i + 2e_j > 1$. Then $p_1p_2^2 > 2x$ for large x, so the cofactor, which has no prime factor below p_2, is 1, and $n = p_1p_2 > x$ with $p_1, p_2 < z_{\text{top}}$. Both primes then lie in $(z_{\text{top}}/4, z_{\text{top}}]$, so there are at most $(12\sqrt{x}/\log x)^2$ such integers for large x, and the corner cells together contribute $O(x/\log^2 x) = o(X)$. □

5.5 The certificate's total

The certificate prices each cell that may be decomposed by what it costs, dropped or decomposed with the terms that its decomposition drops, and takes the cheaper of the two, the second only where the rule allows it.

- A counted cell of depth six costs Δ_6, or, decomposed, the sum of the bounds Δ_8 of the counted cells of depth eight below its cells of depth seven to which (O) does not apply.

- A counted cell of depth four costs Δ_4, or, decomposed, the sum of the costs of the counted cells of depth six below its cells of depth five to which (E) and (O) do not apply.

- A candidate cell (i, j) costs $\Delta_2(i, j)$, or, decomposed, $\Delta_2^{\mathrm{dec}}(i, j)$, the sum over its cells of depth three to which (E) and (O) do not apply of the costs of their counted cells of depth four, rounded up to an integer, as Δ_2 is.

- The total is the sum of the costs of the candidate cells, $\Sigma = \sum_{i,j} \mathrm{cost}(i, j)$.

The certificate checks the rule on the boxes of depth three and more through a sufficient condition, the *block test*, which on the boxes of depth three is the rule itself. For the box (i, j, k) of the first three primes it computes the least sum $\Delta_{ijk}^{\mathrm{blk}}$ of the upper ends of deeper cells at which no split of the box, with the deeper cells joined to one of its sides, passes the bilinear test with the box's own T; a deeper box passes the block test if its upper sum is below $1 - U_1/D$, or if it has no boundary term and the upper ends of its deeper cells sum to less than $\Delta_{ijk}^{\mathrm{blk}}$. A box that passes the block test passes the rule, since joining cells to a box can only lower its T, and the bilinear test's inequalities are monotone. In the same way the certificate finds the surely evaluable boxes of depths four to six from a table over the sums of the cells' indices, which is sound since $h_C = 4h$ makes the sums $s^{\pm}$ of a box functions of its numbers of fine and of coarse cells and of the sums of their indices (a coarse cell's index counting four). The certificate consults the table at depth five, and at depth six only for the cells it does not decompose; below a counted cell of depth four whose deeper boxes all pass the block test it does not consult it, and charges every cell of depths five and six of the region, so that the cell's cost depends only on its index and on the sum of the indices above it. Charging a cell that (E) evaluates only raises the total. So the certificate's decisions are among those the rule allows, the boxes it evaluates are surely evaluable, and its total bounds the dropped terms of its own decomposition; these devices make its computation feasible.

On the grid of Proposition 5.4, the certificate decomposes 15884 of the 23173 candidate cells and drops the other 7289: the rule fails on the box of 1725 of them, and on a box of depth three below 5542 others, and the last 22 would cost more decomposed. The decomposed cells cost $0.2149\mathcal{K}$ in all, the cost of the cells dropped below them: $0.1572\mathcal{K}$ at depth four, $0.0083\mathcal{K}$ at depth six and $0.0494\mathcal{K}$ at depth eight (computations, not formalised, printed

by `tools/linnik/breakdown8.py`). Figure 1 shows the kinds of candidate cell.

The other cells of depth two cost nothing in the total: they are evaluated, empty, or corner cells (Lemma 5.2(3)). With the decomposition stopped at depth six, that is with no counted cell of depth six decomposed, the total Σ would be about $1.55502\mathcal{K}$ on the grid of Proposition 5.4, and stopped at depth four about $6.36647\mathcal{K}$; with (R1) alone as the rule, that is without the bilinear test, it would be about $2.43064\mathcal{K}$, and $2.77411\mathcal{K}$ stopped at depth six (computations, not formalised, by the variants of `tools/linnik/breakdown8.py`). So what lets the certificate close at $11/5$ is the bilinear test together with the two deepest levels of the decomposition.

Proposition 5.3. *Assume the conditions of Theorem 6.1 on L, the window and the grid, and the conclusion of Theorem 3.1. If $\Sigma/\mathcal{K} \le c$, then for every $\varepsilon > 0$, for all large q and every a prime to q,*

$$S(\mathcal{A}, z_{\mathrm{top}}) \ge \frac{1}{\varphi(q)} S(\mathcal{B}, z_{\mathrm{top}}) - (c+\varepsilon)\frac{x}{\varphi(q)\log x}.$$

Proof. The identities. Buchstab's identity, the division of each prime's range into cells and the splits at X' are exact, so $S(\mathcal{A}, z_{\mathrm{top}})$ is the signed sum of the terms of the tree, and so is $S(\mathcal{B}, z_{\mathrm{top}})$. Compare the two term by term. A dropped term has sign $+$ and an $\mathcal{A}$-count at least 0. Every other term is evaluated or empty, or is decomposed. A decomposed term is the signed sum of the terms below it, so one with sign $-$ is bounded from above, and one with sign $+$ from below, by bounding those terms: the decomposed terms of sign $-$ are those of depths one, three, five and seven, and those of sign $+$ the candidate cells of depth two and the counted cells of depths four and six that are decomposed.

The errors. Let $\mathrm{Err} = \varepsilon_H x/(\varphi(q)\log q) + q^{L-\delta}/\varphi(q)$, with the ε_H of Propositions 4.7 and 4.8 small and the least of the δ of Lemmas 4.5 and 4.6 and Propositions 4.7 and 4.8, which are applied to finitely many families of comparisons, since the boxes are finitely many. Every evaluated term and every term $S(\cdot, z_0)$ has its $\mathcal{A}$- and $\mathcal{B}/\varphi(q)$-counts within Err, and there are at most C_{dec} terms, a number that depends on N alone. So

$$S(\mathcal{A}, z_{\mathrm{top}}) \ge \frac{S(\mathcal{B}, z_{\mathrm{top}}) - \mathrm{Dr}(\mathcal{B})}{\varphi(q)} - C_{\mathrm{dec}}\,\mathrm{Err},$$

where $\mathrm{Dr}(\mathcal{B})$ is the sum of the $\mathcal{B}$-versions of the dropped terms. Taking ε_H small in terms of ε and q large, the errors are below $\varepsilon x/(2\varphi(q)\log x)$.

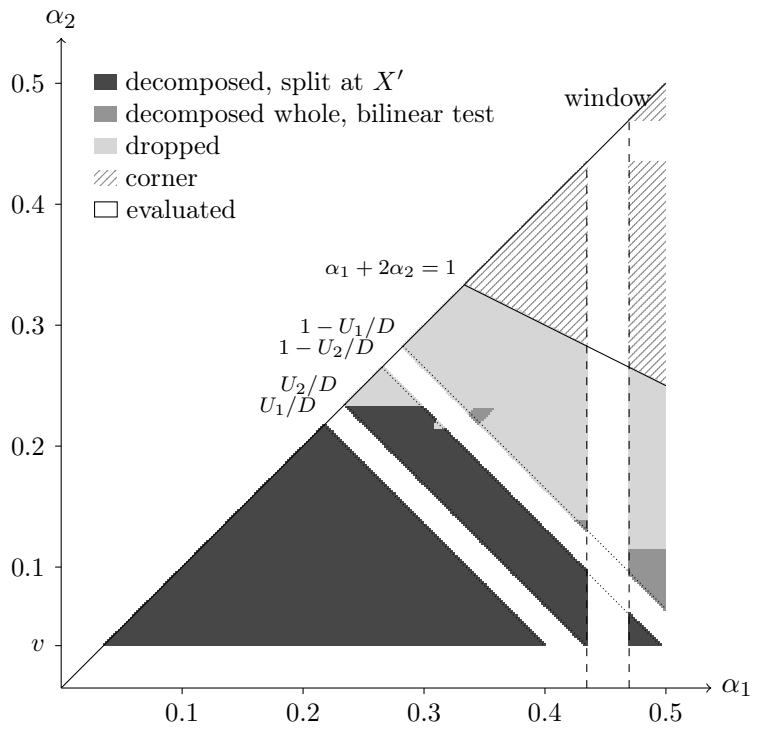


Figure 1: The cells (α_1, α_2) of depth two on the grid of Proposition 5.4, with $v \le \alpha_2 < \alpha_1 < 1/2$. Shaded from dark to light, the three kinds of candidate cell: the 15421 that are decomposed and split at X', which cost $0.1892\mathcal{K}$ in all; the 463 decomposed whole, by the bilinear test, $0.0257\mathcal{K}$; and the 7289 dropped, $0.7173\mathcal{K}$. Of the dropped cells' cost, $0.6790\mathcal{K}$ is in the 6896 that reach the line $\alpha_1 + \alpha_2 = 1 - U_1/D$ (Section 8). White: the cells evaluated at depth two, on which $\alpha_1 + \alpha_2$ or α_2 lies in $\mathcal{U}$ throughout the cell, and the cells of the window's strip $U_1/D < \alpha_1 < U_2/D$, between the dashed lines, where the term of depth one is evaluated; a cell that straddles a dashed line is decomposed at depth one. The dotted lines are $\alpha_1 + \alpha_2 = U_1/D$, U_2/D, $1 - U_2/D$ and $1 - U_1/D$, labelled by their constants. Hatched: the corner cells, beyond $\alpha_1 + 2\alpha_2 = 1$. Drawn by `tools/linnik/fig_cells.py` from the certificate's definitions.

The dropped terms. By Lemma 5.2, a dropped candidate cell contributes at most $(\Delta_2(i,j)/\mathcal{K}+\varepsilon_c)X$ to $\mathrm{Dr}(\mathcal{B})$, and a decomposed candidate cell at most $(\Delta_2^{\mathrm{dec}}(i,j)/\mathcal{K}+M\varepsilon_c)X$, with $M=1+N^2(1+N_C^2(N_C^2+1))$ bounding the number of its dropped cells of depths four, six and eight; the corner cells contribute $o(X)$. So each candidate cell contributes at most $(\mathrm{cost}(i,j)/\mathcal{K}+M\varepsilon_c)X$. Adding up and taking ε_c small, $\mathrm{Dr}(\mathcal{B})\le(c+\varepsilon/2)X$ for large q. $\square$

5.6 The certificate

Proposition 5.4. *On the grid of $N=272$ cells a side, with $N_C=68$ cells of four a side in the coarse grid, and with $U_1=\lceil(43457/100000)D\rceil$ and $U_2=\lfloor(46959/100000)D\rfloor$, under the rule at $L=11/5$, $\Sigma<\mathcal{K}$.*

The total is a finite sum of rational expressions in natural numbers, rounded to integers, so the proposition is a computation. The formalisation checks it by evaluation in Lean's kernel, in 1136 parts, each bounded by 1.001 times its value, rounded up; their bounds sum to 1075756859466551505, that is $0.9330703\ldots\mathcal{K}$, and the exact value is $\Sigma=1074682177289261666$, that is $0.9321382\ldots\mathcal{K}$ (Section 7 and §S5).

6 Proof of Theorem 1.1

The argument of Sections 4 and 5 uses of the particular values of L, u_1, u_2 and N only the conditions of the following theorem, of which Theorem 1.1 is a case.

Theorem 6.1 (the route theorem)**.** *Let $2<L\le 5/2$ and $0<u_1<u_2<1$, and let a grid of N cells a side, with $4\mid N$, integer ends U_1 and U_2 over $D=2^{40}$ and its coarse grid of $N/4$ cells of side $h_C=4h$, satisfy $u_1D<U_1$, $U_2<u_2D$ and $LU_2>D$, and the grid conditions*

$$2U_2<D,\quad 2e_{N-1}<1<2(v+Nh),\quad 20h\le v.$$

Assume the conclusion of Theorem 3.1 for L on the window $[u_1L,u_2L]$, and $\Sigma<\mathcal{K}$ on the grid, under the rule at L. Then there is K' such that for every $q\ge1$ and every a prime to q there is a prime $p\equiv a \pmod q$ with $p\le K'q^L$.

The grid conditions are what the proofs of Sections 5 and 6 use of the grid. $2e_{N-1}<1<2(v+Nh)$ gives $h>0$, and then $20h\le v$ gives $v>0$, that is $U_1<U_2$; so the cells and the floor $z_0=\lceil x^v\rceil$ are well defined. $1<2(v+Nh)$ makes the cells, and the coarse cells, cover the exponents of the primes below

z_{top}. $2e_{N-1} < 1$ gives $2e_i < 1$ for every cell, hence $v < 1/2$ and $z_0 \le z_{\mathrm{top}}$, and, since $h < e_{N-1}$, $v + Nh = e_{N-1} + h < 1$, which keeps the thresholds of the comparisons of every cell, the coarse ones included, in $[1, q^{L+1}]$. $2U_2 < D$ gives $z_{\mathrm{top}} \le X'$. That $4 \mid N$ makes a coarse cell four cells of the grid, of side $h_C = 4h$, and the sums $s^{\pm}$ of a box functions of its numbers of fine and of coarse cells and of the sums of their indices, as the certificate's table of the surely evaluable boxes needs. $20h \le v$ gives $y \ge 1$ in Lemma 5.2 ($4h \le v$ at depths two and four, $12h \le v$ at depth six, $20h \le v$ at depth eight). Of the other conditions, $LU_2 > D$ gives $b_0 < L - 1$, and with $L \le 5/2$ it gives $X'^{3/2} < N_1$ for the empty terms; $L \le 5/2$ also bounds the number of pieces in the proof of Lemma 4.5. The condition $2 < L$ follows from $LU_2 > D$ and $2U_2 < D$; Lemma 4.3 uses it, through Proposition 4.8, and the rest of the argument uses only $L > 1$.

Proof. Let $\rho = \Sigma/\mathcal{K} < 1$ and $0 < \varepsilon < (1 - \rho)/4$. By the prime number theorem, for large q the primes in $(x, 2x]$ that do not divide q number at least $(1 - \varepsilon)x/\log x$, and each of them is counted by $S(\mathcal{B}, z_{\mathrm{top}})$. Proposition 5.3 gives, for large q and every a prime to q,

$$S(\mathcal{A}, z_{\mathrm{top}}) \ge (1 - \varepsilon - \rho - \varepsilon)\frac{x}{\varphi(q)\log x} > \frac{1-\rho}{2} \cdot \frac{x}{\varphi(q)\log x} > 0.$$

An integer of $\mathcal{A}$ counted by $S(\mathcal{A}, z_{\mathrm{top}})$ is at most N_2 and has no prime factor at most $N_2^{1/2}$, so it is a prime $p \equiv a \pmod q$ with $x < p \le 2q^L$. So there is q_0 such that the theorem holds for $q \ge q_0$ with $K' = 2$. Each of the finitely many pairs $(q, a \bmod q)$ with $q < q_0$ has a prime $p \equiv a \pmod q$ by Dirichlet's theorem, and K' is taken large enough to cover them. □

Proof of Theorem 1.1. Take $L = 11/5$, $u_1 = 43457/100000$, $u_2 = 46959/100000$ and the grid of Proposition 5.4. Its ends are rounded inward, $11U_2 > 5D$, and its grid conditions hold, by a computation in integers. Theorem 3.1, under Inputs 1 and 2, is the Type II estimate, and Proposition 5.4 is $\Sigma < \mathcal{K}$. Theorem 6.1 gives Theorem 1.1 under the two inputs. Input 1 enters through Lemma 3.2 and the top bin in Section 3, through the kernel bound of Section 3.3 (where the Pólya–Vinogradov inequality would do instead), through the piece of g^* above $11/12$, which only makes g^* a density curve on all of $[1/2, 1]$, and, in the formalisation, through the prime number theorem of Sections 4 to 6, which it derives from the case of ζ in the form $\psi(t) - t = O(t^{11/12+\varepsilon})$. □

The proof does not compute K: the threshold q_0 comes from limits in q in each step, and the small moduli from Dirichlet's theorem. As in the classical

proofs, every constant is effective in principle; unlike them, the argument needs no step on an exceptional zero, since Input 1 leaves none. But its thresholds are enormous: the inward rounding of the window to multiples of 2^{-40}, for one, leaves Proposition 4.7 margins below 10^{-12} in its exponents, which only $\log q \gtrsim 10^{12}$ turns into a usable factor.

7 The formalisation

Appendix A names the Lean theorems that prove each numbered result of Sections 1 to 6, and the results from the literature that the proof uses; they are in `lib/` of the repository [49] or in Mathlib. `formalisations/` documents the results from the literature and the further classical results behind them. The Lean statement of Theorem 1.1 is

```
theorem linnik_22000e : ∃ K : ℝ, ∀ (q : ℕ) [NeZero q] (a : ℕ),
    a.Coprime q → ∃ p : ℕ, p.Prime ∧ p ≡ a [MOD q] ∧
      (p : ℝ) ≤ K * (q : ℝ) ^ (11 / 5 : ℝ)
```

where `[NeZero q]` says $q \geq 1$ and the power is the real power. The file `lib/check/Check.lean` prints the axioms of the library's theorems in Appendix A, and `scripts/check_axioms.py` checks that they are `propext`, `Classical.choice` and `Quot.sound` only. The axioms of a theorem cover its whole dependency cone, so a `sorry` anywhere in it would appear as `sorryAx`; the library's own sources contain no `sorry`, no declared axiom and no `native_decide` (`scripts/check_hygiene.py`). The certificate of Proposition 5.4 is checked by kernel evaluation, `decide +kernel`, and not by compiled code, in 1136 parts, on definitions that are arithmetic on Lean's core natural numbers, lists and booleans. The toolchain is Lean `v4.35.0-rc3` with Mathlib at commit `0f64d30a`, and the repository's continuous integration builds the library and checks the axioms on every change to it.

Among the statements that are not Lean theorems are: the historical statements of Sections 1 and 8, among them the remarks on R. Li's and on Banks and Shparlinski's results; the heuristic remarks of Sections 1, 3 and 4, among them what the window needs of the kernel bound, and the effectivity remark of Section 6; the remarks that Chen, Gupta and Li's Theorem 1.2 and Corollary 1.4 give $7/3+\varepsilon$; the account of where the window comes from and of its losses (Sections 3.5 and 3.6); the figures computed by the repository's scripts (the exact total Σ and the totals 1.55502, 6.36647, 2.43064 and 2.77411 times $\mathcal{K}$ of Section 5, the counts and costs of the cells there and in Figure 1, the slacks, losses and sensitivities of the window, among them the worst case of the A_k, and the computations of Section 8); and what is said about the sources of the inputs and their independent checking.

Independent checking of the inputs is limited: OpenAI's own Lean formalisation of the main preprint's result [11] is OpenAI's record, not an independent one; this library's proof of Input 1 is the author's new formalisation of the companion preprint's route; and for Input 2 the author knows of no independent check. §S5 gives the details: what of Chen, Gupta and Li's paper is formalised, the Lean statements of the density estimates, the Lean forms of the paper's definitions and checks, the library's proof of Input 1, independent checking, what the kernel checks, and the artefact.

8 What the inputs give without the sieve, and the method's limits

This section claims nothing new. It places Theorem 1.1 against what the same inputs give by the classical argument, which has no sieve, and against the exponents known for special moduli; the table of the introduction sums it up.

The classical deduction

Suppose that for some $A \geq 2$ there is a uniform density estimate

$$N^*(\sigma, T; q) \leq C(qT)^{A(1-\sigma)}(\log qT)^B \qquad (1/2 \leq \sigma \leq 1,\ T \geq 2).$$

With Input 1 it gives Linnik's theorem with every exponent $L > A$. The library's proof, `Linnik.linnik_of_charDensity`, writes the smoothed sum $\sum_{n\equiv a} \Lambda(n)e^{-n/x}$ through the characters, bounds each non-trivial character's sum by a contour shift to the edge of its largest zero-free rectangle, at one of the levels $\theta_j = \min(1/2 + j\delta/2, 11/12)$, for a small $\delta > 0$ fixed in terms of $L - A$, and counts the characters at each level by the density estimate at the height $T + 1$, with $T = 2\log x + 2$: the characters with zeros of real part about σ contribute about $(q^A/x)^{1-\sigma}$ together, and none has a zero of real part above $11/12$. With it the library proves:

- $L = 12 + \varepsilon$ from Input 1 alone: at most $\varphi(q) \leq qT$ characters are counted for every σ, and at most one above $11/12$, which is the estimate with $A = 12$;
- $L = 36/13 + \varepsilon$ with Ingham's estimate, whose exponent $3/(2 - \sigma)$ is $36/13$ at $\sigma = 11/12$;
- $L = 12/5 + \varepsilon$ with Huxley's estimate (with Ingham's below $\sigma = 3/4$), which bounds the density exponent by $12/5$ on the whole range;

- $L = 2.3334 + \varepsilon$ with the curve g^*, since $g^*(\sigma) \le 2.3334(1-\sigma)$ on $[1/2, 1]$: on every step the exponent is at most $A_0 = 2.3334$, and $g^* = 1 \le 2.3334(1-\sigma)$ for $\sigma < 4/7$, as $2.3334 \cdot 3/7 > 1$.

Appendix A names the Lean theorem of each. With Chen, Gupta and Li's estimate itself the exponent is $7/3 + \varepsilon$, by the same deduction from their Theorem 1.2 [12] (exponent $A = 7/3$), which the library does not formalise as they state it. They do not state a Linnik exponent, but their Corollary 1.4 [12], on primes in progressions in short intervals under a zero-free region, contains this one: for $h \le x$ it gives $\psi(x+h; q, a) - \psi(x; q, a) \sim h/\varphi(q)$ whenever $h/\varphi(q) \ge x^{1/2+\varepsilon} q^{1/6} + x^{17/30+\varepsilon}$, and with $h = x$ and Input 1 as the zero-free region this holds at $x = q^L$ for every $L > 7/3$ and all large q.

Special moduli

For special moduli smaller exponents are known. For the powers of a fixed prime, Banks and Shparlinski [43] have 2.1115 (the value as cited in [12, footnote 1]). R. Li [42] adapts his Harman-sieve calculation to primes in progressions to large moduli q whose prime factors are all below q^C, for an absolute constant $C > 0$ whose existence his theorem asserts without giving it, when the L-functions of the primitive characters to the divisors of q have no zero with $\operatorname{Re} s > 1 - (\log q)^{-3/4}$ and $|\operatorname{Im} s| \le \exp(\varepsilon (\log q)^{3/4})$, with the $\varepsilon > 0$ of his Theorem 9; for the powers of a fixed prime he deduces the exponent $1/0.4752 < 2.1044$ (his Theorem 11), improving Banks and Shparlinski's. He presents these theorems as modifications of that calculation, without separate proofs. Input 1 implies the zero-free hypothesis of his Theorem 9 for every large q, so if that theorem holds as stated, it gives with Input 1 the exponent $1/0.4752$ for every large modulus whose prime factors are all below q^C.

What the half-plane is used for

Neither the classical deduction nor Theorem 1.1 needs the value $11/12$ of Input 1. The deduction works with any zero-free half-plane $\operatorname{Re} s > 1 - \eta_0$, for a fixed $\eta_0 > 0$, in place of Input 1. In the sieve, the half-plane enters the window condition only at the top of the curve: with Input 1 replaced by such a half-plane, the top piece $(101/50)(1-\sigma)$ of g^* extended up to $1 - \eta_0$, and the margins of Definition 3.3 kept, the window of Theorem 3.1 stays admissible at $11/5$ for every $\eta_0 \ge 2.85 \cdot 10^{-4}$. The binding corner is then E_1 at $(1 - \eta_0, \mu_2)$; the corner of E_2 at $\mu = 1$ needs only $\eta_0 \ge 2.4 \cdot 10^{-4}$, and with smaller margins any fixed $\eta_0 > 0$ would do. The other uses of Input 1 also

need only some half-plane: the contour of Lemma 3.2, the prime number theorem, and the kernel bound, for which the Pólya–Vinogradov inequality would do (Section 3.6). So Theorem 1.1 depends on OpenAI's result only through the existence of some zero-free half-plane, that is, no zeros near $\operatorname{Re} s = 1$ and in particular no exceptional zero. This is a remark, checked at the corners in exact rationals, not a theorem of the library, whose statements use $\operatorname{Re} s > 11/12$.

Where the gain comes from, and its limits

The deduction needs the zeros of every real part to contribute less than the main term, so its exponent is the largest density exponent on $[1/2, 11/12]$, which is $7/3$, at $\sigma = 5/7$. The sieve needs the zeros only in bilinear sums with one variable in the window, where Theorem 3.1 trades the density exponent against the length of the variable, and recovers the primes from those sums at the cost that the certificate bounds; the window is set where the density exponent is largest (Section 3.5). With steps of Ingham's and Huxley's estimates, of length $1/1680$, in place of Input 2 on $[5/7, 131/168]$, the window is set at $3/4$, where both exponents are $12/5$, and a window admissible at $L = 2239/1000$ carries a certificate's total below $\mathcal{K}$ on 272 cells a side, a grid that satisfies the conditions of Theorem 6.1 (at 2.238 the window is too narrow for that grid, and finer grids were not tried), so the sieve brings $12/5$ down to about 2.24, with Input 1 as the only input that is not refereed (a computation, §S6).

The certificate at $11/5$ has room: with the window chosen afresh, the same method reaches about 2.198, where the certificate's total on a finer grid reaches $\mathcal{K}$ (§S6), and $11/5$ is the round exponent the paper proves. About three quarters of the certificate's total comes from the cells that reach the line $\alpha_1 + \alpha_2 = 1 - U_1/D$ (Figure 1). Most of their integers have no product of some of their prime factors with its exponent in $\mathcal{U}$, so that the Type II estimate cannot count them, and the decomposition reaches them only through the bilinear Type I estimate, where its test passes. More depth can lower only the cost of the decomposed cells, about a quarter of the total, and the half-plane $\operatorname{Re} s > 7/8$ does not help (§S6). Going much lower needs a wider window, that is a better density exponent near $\sigma = 5/7$, or Type I information on longer products. On the former, Chen, Gupta and Li note that improving the term of their estimate that dominates at $\sigma = 5/7$ would need a joint cancellation which "seems quite difficult" [12, Remark 12.3].

Acknowledgements

This work was carried out with the assistance of an AI system, Claude, developed by Anthropic. It was used to develop the argument, to write the Lean formalisation, including the proofs of the results from the literature, and to draft this paper. The author is responsible for the paper's content.

Grok Bot, an AI research assistant built by xAI, also assisted: with the literature and the analysis of the density bounds, with the construction of the Chen–Gupta–Li step curve and the certificate pipeline, with edits to this paper, and with a survey and refactoring of the Lean library for duplication and build cost. The Lean verification is by the Lean kernel and the repository's continuous integration, and not by the assistants.

A The claims table

The table names the Lean theorem behind each result of the paper, and behind the results from the literature that its argument uses directly, with the main ones behind the inputs and the density estimates. Of the 30 steps of Input 2, and of Ingham's and Huxley's steps, `lib/check/Check.lean` prints the first and the last; `SieveCurve7D.counts_7d` uses every step of the curve, so all of them are in the dependency cone of `Linnik22000E.linnik_22000e`, whose axioms are checked. `lib/check/CheckAll.lean` prints the axioms of further declarations of the library.

A name's first component is its namespace, which is usually also the name of its file in `lib/pnt`. The exceptions in the table are these, in `lib/pnt` except the last four, which are in `lib/core`; Dirichlet's theorem is Mathlib's.

`SieveCert.gridOK22000E`	`SieveGrid22000E.lean`
`SieveCert.gridOKC22000E`	`SieveGrid22000E.lean`
`SieveDeep.totalD_lt_22000E`	`SieveCert22000E.lean`
`SieveDeep.Ch3_le`	`SieveDeepCost.lean`
`SieveBuch.Par.decomp8`	`SieveBuch8.lean`
`SieveBuch.Par.I_S6c`, ..., `SieveBuch.Par.I_S8`	`SieveBuch8.lean`
`SieveBuch.Par.I_S7c_vanish_lt`, ..., `SieveBuch.Par.I_S8c_vanish_reg`	`SieveBuch8.lean`
`SieveCert.ibOK_sound`	`SieveRule.lean`
`SieveCert.okIB_cases`	`SieveRule.lean`
`SieveRough.evLe_box_tr`	`SieveBox.lean`
`SieveRough.evLe_box_ex`	`SieveBox.lean`
`SieveRough.shareBox`	`SieveBox.lean`
`HalfPlanePNT.psi_isBigO_eleven_twelfths`	`HalfPlaneFinal.lean`
`SieveTypeIIHM.sum_bin_hm`	`SieveHMBin.lean`
`SieveTypeIIHM.sum_typeII_bins_hm`	`SieveHMBins.lean`
`SieveTypeIIHM.offK_bound`	`SieveHMBins.lean`
`SieveCurveGlue.WindowOKHM`	`SieveHMWindow.lean`
`Linnik.linnik_of_charDensity`	`LinnikDeduction.lean`
`Linnik.linnik_twelve`	`LinnikDeduction.lean`
`Eis.LFunction_ne_zero_of_re_gt`	`DirichletCharHalfPlane.lean`
`Eis.kubotaTheta`	`KubotaThetaProof.lean`
`Eis.quadLargeSieve`	`EisensteinQuadSieveFinal.lean`
`Eis.cub_recip`	`EisensteinCubicRecip.lean`

Claim	Lean
Theorem 1.1	`Linnik22000E.linnik_22000e`
Definition 2.1, density curves	`SieveTypeIIGen.CurveCounts`
Proposition 2.2: g^* is a density curve	`SieveCurve7D.counts_7d`
and $g^* \le 1$, $g^*(1/2) = 1$	`SieveCurve7D.g7d_le_one`, `SieveCurve7D.g7d_half`
Theorem 3.1, the Type II estimate	`SieveTypeIIU22000D.typeII_22000d`
Lemma 3.2, prime sums in a zero-free rectangle	`SieveWeight.norm_Sp_le`
Orthogonality	`SieveTypeII.sum_chars_dp_sq_le`
The Halász–Montgomery inequality in a bin	`SieveTypeIIHM.sum_bin_hm`
The bins, with the better of the two bounds in each	`SieveTypeIIHM.sum_typeII_bins_hm`
The kernel bound	`SieveTypeIIHM.offK_bound`, `LinnikOffDiag.norm_offK_le`, `LinnikLindelof.lindelof`
Definition 3.3, admissible windows	`SieveCurveGlue.WindowOKHM`
Lemma 3.4, one bin	`SieveTypeIIHM.bin_bound_gen_hm`

Claim	Lean
Proposition 3.5, the Type II estimate on an admissible window	`SieveTypeIIHM.typeII_gen_hm`
Proposition 3.6, the window is admissible	`SieveTypeIIU22000D.window_22000d`
Lemma 4.1, Type I counts	`SieveCount.card_A`, `SieveCount.card_B`
Lemma 4.2, the bilinear Type I estimate	`SieveTypeIB.typeIB_bound`
with a spline weight	`SieveSpline.typeIB_spline`
Lemma 4.3, the leaves by the bilinear estimate	`SieveP6Leaves.leaves_chain`, `SieveTypeIBAsym.leaves_ib`
Lemma 4.4, Type II families	`SieveFamily.typeII_sharp`
Lemma 4.5, sharp prime ranges	`SieveTypeIIU.typeII_unw`
Lemma 4.6, chains of primes	`SieveChain.chainTU_bound` (a), `SieveChain.chainTL_bound` (b), `SieveChainR.chainTR_bound` (c), `SieveChain.sum_rough_tuples` (cofactor)
Proposition 4.7, Harman's comparison in the form used	`SieveProp6.prop6`
its tree identity	`SieveTree.tree_pt`
the stops	`SieveStops2.stops2_bound`
a leaf (§S3)	`SieveProp6.leaf_bound`, `SieveLeaf.leaf_gap`
Brun's sieve	`SieveBrun.fundamental`
the main term at a leaf, with $\prod_{p<w}(1-1/p) \le 1/\log w$	`SieveLeaf.prod_T_le`, `SieveLeaf.prod_primesBelow_le`
Proposition 4.8, Harman's comparison with the bilinear estimate	`SieveP6ib.prop6ib`
its tree with a stopping rule	`SieveTreeStop.tree_pt_stop`, `SieveP6Tree.tree_chains`
its leaves	`SieveP6TreeLeaves.tree_leaves_le`
The rule, (R1) or (R2), and Proposition 4.8's hypotheses from (R2)	`SieveCert.okIB_cases`, `SieveCert.ibOK_sound`, `SieveQ.P6IB_box`
Buchstab's identity	`SieveDecomp.buchstab_fam`
The decomposition's identity	`SieveBuch.Par.T_eq`, `SieveBuch.Par.I_S1`, ..., `SieveBuch.Par.I_S6`, `SieveBuch.Par.I_S6c`, ..., `SieveBuch.Par.I_S8`
its empty terms	`SieveBuch.Par.I_S3c_vanish`, ..., `SieveBuch.Par.I_S8c_vanish_reg`
The evaluated terms of the decomposition	`SieveQ8.leaves_ev8`

Claim	Lean
Lemma 5.1, rough numbers from above	`SieveRough.psi_le`
Buchstab's inequality for rough numbers	`SieveRough.Psi_le_buchstab`
its delay inequality	`SieveRough.super_M`
the bound 0.5671433 for the maximum of Buchstab's ω on $[2,3]$	`SieveOmegaB.omB_le_w1`
The bound Ω for ω^* on a cell, with the table	`SieveCert.omegaC_le_w`,
	`SieveOmegaB.omF_ge`,
	`SieveOmegaB.lnT_ge`
the increase of ω on $[2,3]$ up to 2.76 and its decrease from 2.77	`SieveOmegaB.omB_mono`,
	`SieveOmegaB.omB_anti`
The cells' prime sums, $t(e,h)$ and $s(e,h)$	`SieveRough.evLe_cell_tr`,
	`SieveRough.evLe_cell_ex`,
	`SieveRough.evLe_box_tr`,
	`SieveRough.evLe_box_ex`,
	`SieveOmegaB.log_le_half_sub_inv`
The bounds Δ_2, and Δ_4, Δ_6, Δ_8 in fixed point	`SieveCert.bound2_ge`,
	`SieveDrop8.trB_le_trapF`,
	`SieveDrop8.exB_le_lastF`
Lemma 5.2, the dropped terms, (1) and (2)	`SieveDrop.drop2_le`,
	`SieveDrop8.drop4_le'`,
	`SieveDrop8.drop6_le'`,
	`SieveDrop8.drop8_le`,
	`SieveRough.share2_of_cand`,
	`SieveRough.shareBox`,
	`SieveDrop8.share4D`, ..., `SieveDrop8.share8D`
(3), the corner cells	`SieveDrop.corner_le`,
	`SieveDrop.corner_ev`
The block test, and the table of the surely evaluable boxes	`SieveDeep.okIB_of_okD`,
	`SieveDeep.surelyEval4_of_lmask`,
	`SieveDeep.surelyEval5_of_emask`,
	`SieveDeep.surelyEval6_of_emask`
The certificate's decisions pass the rule	`SieveDeep.scheme`
The certificate's costs bound the dropped cells	`SieveDeep.Ch3_le`,
	`SieveDrop8.dT_le_cost`

Claim	Lean
Proposition 5.3, the decomposition (assembled in `SieveMain8.linnik_large8`)	`SieveBuch.Par.decomp8`,
	`SieveQ8.leaves_ev8`,
	`SieveDrop8.drop_ev8`
Proposition 5.4, the certificate	`SieveDeep.totalD_lt_22000E`
Theorem 6.1, the route theorem	`SieveMain8.linnik_route_lt8`
its grid conditions for the grid of Proposition 5.4	`SieveCert.gridOK22000E`,
	`SieveCert.gridOKC22000E`,
	`Linnik22000E.deepGrid22000E`
the other conditions on L, the window and the grid	`Linnik22000E.routeOK22000D`,
	`Linnik22000E.fits22000E`,
	`Linnik22000E.ibFits22000E`
for large q	`SieveMain8.linnik_large8`
Dirichlet's theorem, for the small moduli (Mathlib)	`Nat.forall_exists_prime_gt_and_modEq`
the prime number theorem, from Input 1	`HalfPlanePNT.psi_isBigO_eleven_twelfths`
Input 1 (OpenAI; `formalisations/`)	`Eis.LFunction_ne_zero_of_re_gt`
Input 2 (Chen–Gupta–Li, their Proposition 3.1 and the steps derived from it; `formalisations/`)	`LinnikCGLSteps.charDensity_s0`,
	..., `LinnikCGLSteps.charDensity_s29`,
	`LinnikCGLHB101.prop31`
Ingham's steps on $[4/7, 5/7]$	`LinnikStairs7.charDensity_in7_0`,
	..., `LinnikStairs7.charDensity_in7_18`,
	`LinnikCGLSteps.charDensity_inT`
Huxley's steps on $[131/168, 5/6]$	`LinnikStairs7.charDensity_hx7_5`,
	..., `LinnikStairs7.charDensity_hx7_19`
Ingham's density estimate (`formalisations/`)	`LinnikInghamC.charDensity_ingham_c`
Huxley's density estimate (`formalisations/`)	`LinnikHuxStep.charDensity_gen`
Huxley's estimate from 5/6 on (`formalisations/`)	`LinnikStep.charDensity_step`
The Halász–Montgomery inequality (`formalisations/`)	`LinnikPower.hm_duality`
Mean values over the characters mod q [29, 30, 31]	`LinnikHybrid.mean_value_chars`,
	`LinnikHybrid.large_values_chars`
The hybrid fourth moment [18], as Bui and Heath-Brown state it [26]	`LinnikFourth.fourth_moment`

Claim	Lean
Chen–Gupta–Li's Propositions 4.6, 6.1, 8.1 and 11.1	`LinnikCGLMat.prop46`,
	`LinnikCGLS2.cgls2`,
	`LinnikCGLP81.cglp81`,
	`LinnikCGLHB101.cgls3`
and their Theorem 10.1, Heath-Brown's bound [14, 15, 16]	`LinnikCGLHB101.cglt101`
Behind Input 1: Kubota's cubic theta function [32, 33, 34]	`Eis.kubotaTheta`
the quadratic large sieve over $\mathbb{Z}[\omega]$ [35, 36]	`Eis.quadLargeSieve`
cubic reciprocity [37]	`Eis.cub_recip`
The classical deduction (Section 8)	`Linnik.linnik_of_charDensity`
Its exponents $12+\varepsilon$, $36/13+\varepsilon$ and $12/5+\varepsilon$	`Linnik.linnik_twelve`,
	`LinnikIngham.linnik_thirtysix_thirteenths`, `LinnikTwelve.linnik_twelve_fifths`
Its exponent $2.3334+\varepsilon$ from the curve g^*	`LinnikG7.linnik_g7d`,
	`LinnikG7.charDensity_g7d`, `SieveCurve7D.g7d_le_lin`

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